Optimal Transport in linear Independent Component Analysis

Master Thesis (M.Sc.)
in
Quantitative Data Science Methods
Psychometrics, Econometrics and Machine Learning
Faculty of Economics and Social Sciences
at the University of Tübingen
Methods Center

submitted by
Ashutosh Jha
from New Delhi, India
(Matriculation number: 6639615)

Submitted in Tübingen
June 8, 2026

1. Supervisor: Prof. Dr. Joachim Grammig[1]

2. Supervisor: Prof. Dr. Michel Besserve[2,3]

3. Supervisor: Dr. Simon Buchholz[2]

1. Reviewer: Prof. Dr. Joachim Grammig[1]

2. Reviewer: Prof. Dr. Augustin Kelava[4]

[1] Chair of Statistics, Econometrics and Empirical Economics, University of Tübingen

[2] Empirical Inference Group, Max Planck Institute for Intelligent Systems, Tübingen

[3] Institute of Artificial Intelligence, TU Braunschweig

[4] Methods Center, University of Tübingen

Declaration of Academic Integrity

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Ashutosh Jha	Tübingen, 10.06.2026
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Chapter 1

Introduction

Blind source separation, specifically formulated as independent component analysis (ICA), is a fundamental problem in statistics (Jutten & Herault, 1985). The objective of ICA is to recover a set of hidden, statistically independent source signals from an observed set of mixtures. In this thesis, only linear mixtures, wherein the observations are assumed to be generated by an unknown, invertible linear combination of the latent sources are considered.

To achieve this separation, ICA imposes statistical independence among the recovered components. This approach is frequently contrasted with principal component analysis (PCA). While PCA diagonalizes the covariance matrix to ensure that the resulting components are uncorrelated, ICA enforces full statistical independence (Hyvärinen, Karhunen, & Oja, 2001, Chapter 1). This stronger criterion allows ICA to disentangle latent variables that remain mixed under simple decorrelation.

1.1 Existing Methodologies

The standard ICA pipeline begins by centering and whitening the observed mixture (Hyvärinen et al., 2001). Whitening is a necessary preprocessing step that removes linear correlations and scales the marginal variances to unity. Once the data is whitened, the search for independent components is restricted to finding an optimal orthogonal rotation matrix on a space of orthogonal matrices (Absil, Mahony, & Sepulchre, 2008).

Traditional ICA algorithms navigate this rotational search by optimizing contrast functions to identify non-Gaussian projections (Hyvärinen et al., 2001, Chapter 8). These classical methods rely on surrogate information-theoretic measures, such as negentropy approximations or cumulant-based matching, to quantify the non-Gaussianity of the unmixed components (Amari, Cichocki, & Yang, 1996).

While computationally efficient, these parametric approximations evaluate surrogate information-theoretic contrasts or specific statistical moments rather than the full probability density. By projecting complex probability distributions onto a limited set of functional expectations (such as the standard *logcosh* approximation), these methods introduce structural blind spots. For instance, they inherently struggle to resolve the sharp geometric features of discrete or highly multi-modal distributions, effectively smoothing over critical density variations.

More severely, because these proxies rely on expected values, specific non-Gaussian topologies can evaluate to the exact same contrast expectation as a standard Gaussian. This creates flat regions in the objective landscape where the gradient vanishes entirely. Furthermore, the second-order fixed-point solvers typically employed to optimize these surrogates rely on estimating local algorithmic curvature. This expected curvature can degenerate to zero, causing the matrix update steps to become mathematically undefined. Consequently, we find that by failing to map the accurate, geometry of the underlying distributions, these proxy-based methods are prone to algorithmic and optimization failures in complex environments.

1.2 Applications

The ability to blindly separate independent sources without parametric assumptions regarding their underlying distributions has interdisciplinary applications. In neuroscience, ICA serves as the standard clinical technique for isolating and removing high-amplitude biological artifacts, such as ocular and muscular signals, from continuous electroencephalogram (EEG) and magnetoencephalogram (MEG) recordings (Hyvärinen et al., 2001, Chapter 21). In telecommunications, ICA is deployed to separate overlapping transmissions in code-division multiple access (CDMA) systems, while in image processing, it is utilized to extract localized, oriented, and bandpass features from natural images, effectively mimicking the processing functions of the mammalian primary visual cortex (Hyvärinen et al., 2001, Chapters 22-23).

Furthermore, the methodology is increasingly utilized in econometrics and market microstructure analysis. In financial price discovery, ICA facilitates the identification of latent structural shocks, allowing researchers to uniquely resolve the orthogonal ambiguities inherent in standard vector error correction models (Zema & Cordoni, 2025). Across all these distinct domains, the validity of the downstream scientific or economic analysis depends strictly on the reliable extraction of the underlying independent components.

1.3 Main Contribution

The primary methodological contribution of this thesis is the formulation of an optimal transport theory based framework for independent component analysis (OT-ICA). The core objective of the OT-ICA framework reduces to searching the space of orthogonal matrices for the rotation matrix that maximizes the non-Gaussianity of the projected data using metrics from optimal transport as contrasts. We establish the theoretical soundness of this proposed method and empirically validate its performance, demonstrating that it effectively isolates independent components from a linear mixture.

To achieve this, we propose utilizing the Wasserstein distance (Villani, 2003) as the contrast function for the rotational search. [1] Our objective is to find the orthogonal candidates that maximize the Wasserstein distance to a standard normal distribution. Because the Wasserstein distance is a proper metric—unlike the proxy approximations utilized in classical methods and allows for highly efficient empirical estimation in one dimension (Villani, 2003), it successfully evaluates the true global geometry of the distribution.

1.4 Structure of the Thesis

The remainder of this thesis is structured as follows. Chapter 2 establishes the theoretical foundations of linear ICA, information geometry, and optimal transport. Chapter 3 formalizes the OT-ICA framework, and proves that the squared Wasserstein (W_2^2) distance between a linear mixture and a standard Gaussian distribution is strictly upper-bounded by the weighted sum of the individual source distances. Chapter 4 details the algorithmic enhancements required to optimize this metric efficiently on the Orthogonal Group or Orthogonal Matrices space. Chapter 5 presents our empirical evaluation, demonstrating the failure modes of standard proxy optimization and how OT-ICA addresses these limitations. Finally, Chapter 6 concludes the thesis with a summary of findings and directions for future research.

[1] The complete implementation of the OT-ICA framework is publicly available at: `https://gitfront.io/r/ashutosh-jha/4gwwN1sPeeAD/ot-in-linear-ica/`

Chapter 2

Theoretical Foundations

This chapter establishes the foundations necessary for linear independent component analysis using optimal transport. We first formalize the linear mixing model and its identifiability conditions, subsequently detailing the ideas that motivate the framework. Following this, we introduce an information-geometric perspective linking statistical independence to non-Gaussianity, and finally, formalize the principles of optimal transport that govern the proposed optimization metric.

2.1 Linear Independent Component Analysis (ICA)

The foundational assumption of the linear Independent Component Analysis model is that the observed signals are generated by a linear combination of statistically independent latent sources. We formalize this generative model and its required constraints as follows:

Theorem 2.1 The Linear ICA Model and Identifiability, (Comon, 1994). *Let the observed data be denoted by a random vector* $\mathbf{X} \in \mathbb{R}^d$*, and the latent sources by* $\mathbf{S} \in \mathbb{R}^d$*. The linear ICA model assumes:*

a) ***Linear Mixing:*** $\mathbf{X} = \mathbf{A}\mathbf{S}$*, where* $\mathbf{A} \in \mathbb{R}^{d \times d}$ *is an unknown, invertible mixing matrix.*

b) The components $s_1, s_2, \ldots, s_d$ *of* $\mathbf{S}$ *are mutually statistically independent.*

c) At most one of the source components s_i *follows a Gaussian distribution.*

Under these conditions, the mixing matrix $\mathbf{A}$ *can be uniquely identified from the observations* $\mathbf{X}$*, up to a permutation and scaling of its columns.*

Expanded into vector notation, the structural mixing represents the observed variables as

$$\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_d \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1d} \\ a_{21} & a_{22} & \cdots & a_{2d} \\ \vdots & \vdots & \ddots & \vdots \\ a_{d1} & a_{d2} & \cdots & a_{dd} \end{bmatrix} \begin{bmatrix} s_1 \\ s_2 \\ \vdots \\ s_d \end{bmatrix}. \tag{2.1}$$

The core objective of ICA is to estimate an unmixing matrix $\mathbf{B} \approx \mathbf{A}^{-1}$ such that the recovered components $\mathbf{Z} = \mathbf{BX}$ are mutually independent, thereby reconstructing the original sources $\mathbf{S}$ up to an arbitrary permutation and scaling

2.1.1 The Insufficiency of Principal Component Analysis

Principal Component Analysis (PCA) is frequently utilized as a baseline technique to decorrelate observed data. This corresponds to finding the orthogonal matrix $\mathbf{V}^\top$ derived from the eigenvalue decomposition of the data's covariance matrix. However, decorrelating a sample is mathematically insufficient for achieving blind source separation when the underlying data is non-Gaussian.

As noted by MacKay (2003), PCA is fundamentally an L_2-norm variance maximization technique; it is inherently sensitive to variable scaling and heavily influenced by outliers. More critically, PCA relies exclusively on second-order statistics. Once the data is decorrelated and scaled such that $\mathbb{E}[\mathbf{XX}^\top] = \mathbf{I}_d$, the second-order structure becomes invariant to any arbitrary orthogonal rotation. While decorrelation implies strict independence for multivariate Gaussian distributions, higher-order dependencies remain intact for non-Gaussian data. Consequently, ICA must search the space of orthogonal matrices to locate the specific rotation that eliminates these higher-order dependencies and achieves independence for a mixture of non-gaussian sources.

2.1.2 Identifiability and Ambiguities

The non-Gaussianity constraint defined in Theorem 2.1 is a strict requirement. If two or more sources are Gaussian, any orthogonal transformation of those specific sources yields another set of independent Gaussian variables, rendering the mixing matrix $\mathbf{A}$ impossible to determine uniquely.

Furthermore, even when the non-Gaussianity assumption holds, the model has two ambiguities. The first is a scaling ambiguity: because both $\mathbf{A}$ and $\mathbf{S}$ are unknown, any scalar multiplier assigned to a source s_i can be cancelled by dividing the corresponding column of $\mathbf{A}$ by the same scalar. Therefore, the variances of the unobserved sources cannot be inferred. The second is a permutation ambiguity: the model treats the sum of sources symmetrically, meaning the indexing order of the recovered components is arbitrary.

2.1.3 Centering and Whitening

To simplify the optimization landscape, the observed data $\mathbf{X}$ is subjected to a two-step preprocessing phase consisting of centering and whitening (Hyvärinen et al., 2001, Chapter 6). Centering ensures that the data has a zero mean by subtracting the expectation, $\mathbb{E}[\mathbf{X}] = \mathbf{0}$. Given centered data, the covariance matrix simplifies to $\mathrm{Cov}(\mathbf{X}) = \mathbb{E}[\mathbf{X}\mathbf{X}^\top]$. Whitening transforms the observed vector into a new vector $\widetilde{\mathbf{X}}$ whose components are uncorrelated and possess unit variance. This is achieved using the eigenvalue decomposition of the covariance matrix

$$\mathrm{Cov}(\mathbf{X}) = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^\top \tag{2.2}$$

where $\mathbf{V}$ is the orthogonal matrix of eigenvectors and $\mathbf{\Lambda}$ is the diagonal matrix of eigenvalues λ_j. The whitening transformation matrix $\mathbf{Q}$ is constructed by scaling the eigenvectors by the inverse square root of the eigenvalues

$$\mathbf{Q} = \mathbf{\Lambda}^{-1/2}\mathbf{V}^\top. \tag{2.3}$$

Applying this transformation yields the whitened data $\widetilde{\mathbf{X}} = \mathbf{Q}\mathbf{X}$, where the covariance is the identity matrix $\mathbf{I}_d$. After whitening, if we define the unmixing operation on the whitened data as $\widetilde{\mathbf{Z}} = \mathbf{B}\widetilde{\mathbf{X}}$, the covariance of the recovered components is

$$\mathrm{Cov}(\widetilde{\mathbf{Z}}) = \mathbf{B}\mathrm{Cov}(\widetilde{\mathbf{X}})\mathbf{B}^\top = \mathbf{B}\mathbf{I}_d\mathbf{B}^\top = \mathbf{B}\mathbf{B}^\top. \tag{2.4}$$

Because the independent components must have unit variance, we require $\mathbf{B}\mathbf{B}^\top = \mathbf{I}_d$. This reveals that the unmixing matrix $\mathbf{B}$ must be orthogonal, thus reducing the ICA optimization to a search over a space of orthogonal matrices.

2.1.4 Information-Theoretic Intuitions

The identifiability limitation in Theorem 2.1 leads to the translation of the challenge of finding independent components into a distinct optimization objective: maximizing non-Gaussianity. The theoretical justification for this approach is rooted in the Central Limit Theorem.

Theorem 2.2 (Central Limit Theorem Intuition). *Let $s_1, s_2, \ldots, s_d$ be independent random variables. Under general regularity conditions, the distribution of their normalized sum tends toward a Gaussian distribution as d increases (Billingsley, 1995).*

While Theorem 2.2 formally applies to limits of sums, its consequence in ICA is that any observed linear mixture of independent, non-Gaussian sources will appear more Gaussian than the underlying sources themselves. (Hyvärinen et al., 2001, Chapter 1). Therefore, by extracting the unmixed components that are the least Gaussian, the algorithm effectively isolates the original independent signals.

The use of non-Gaussianity as a proxy for independence is further understood from an information-theoretic perspective. For any continuous random variable with a fixed finite variance, the Gaussian distribution uniquely possesses the maximum possible Shannon entropy (Cover & Thomas, 2006, Chapter 8). The Central Limit Theorem dictates that the linear mixing of independent variables yields a distribution that is closer to this maximum-entropy Gaussian state than the original constituent sources (Hyvärinen et al., 2001, Chapter 8). Therefore, the process of unmixing is equivalent to searching for the rotation matrix that reverses this entropy increase, identifying the original sources by locating the projections with the maximum non-Gaussianity.

2.2 An Information Geometry Perspective on ICA

2.2.1 Product and Gaussian Manifolds

Information geometry provides a framework for quantifying the relationship between independence and non-Gaussianity by analyzing probability distributions as points on geometric manifolds. Following the perspective introduced by Cardoso (2022), ICA can be understood as orthogonally projecting the empirical distribution onto two distinct subspaces. The first is the Product manifold $\mathcal{P}$, which contains all distributions where the marginal components are strictly independent. The second is the Gaussian manifold $\mathcal{G}$, which encompasses all multivariate Gaussian distributions with zero mean and a fixed covariance structure. We dive into this perspective with the knowledge that standard decorrelation merely ensures that the covariance is diagonal, but full ICA seeks to find a linear transform that minimizes the divergence between the empirical data and its projection onto the Product manifold (Cardoso, 2022).

2.2.2 Kullback-Leibler Pythagorean Identities

Distances between these distributions are measured using the Kullback-Leibler (KL) divergence (Cover & Thomas, 2006). For any random vector Y with a joint density P_Y, the mutual information $\mathscr{I}(Y)$ measures the distance to the closest

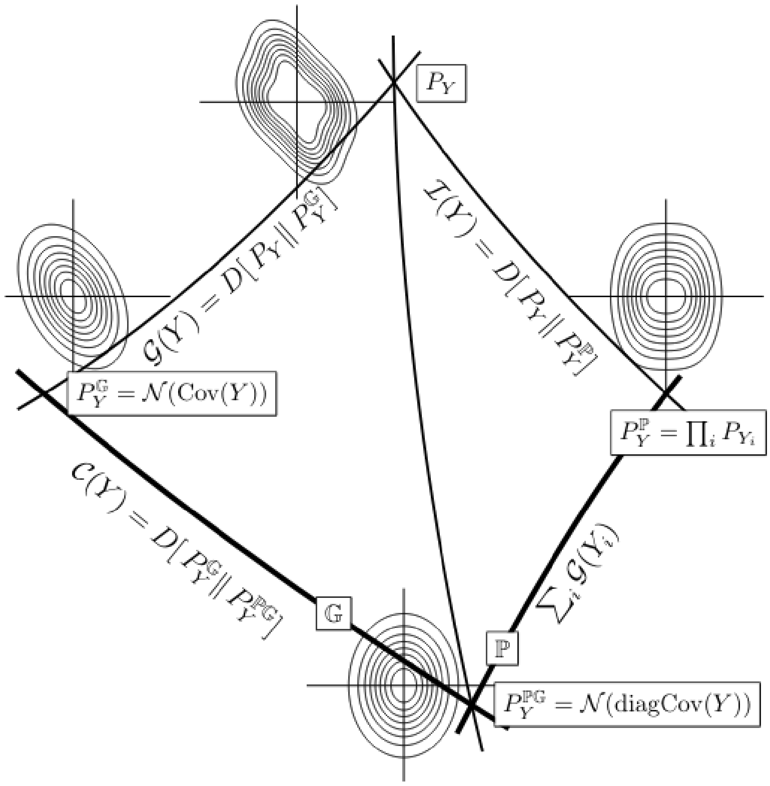

Figure 2.1: *The Pythagorean geometry of Independent Component Analysis. The empirical joint distribution P_Y is orthogonally projected onto the Product manifold $\mathcal{P}$ (yielding the independent marginals P_Y^P) and the Gaussian manifold $\mathcal{G}$ (yielding the best Gaussian fit $\mathcal{N}(\mathrm{Cov}\, Y)$). The intersection of these approximations yields the fully factorized Gaussian target $\mathcal{N}(\mathrm{diag}\,\mathrm{Cov}\, Y)$. The Kullback-Leibler divergences between these nodes geometrically represent distinct statistical properties: Mutual Information $\mathscr{I}(Y)$, Joint Non-Gaussianity $G(Y)$, and Correlation $C(Y)$. Figure from Cardoso (2022).*

independent product distribution, denoted $P_Y^P = \prod_i P_{Y_i}$. This expands as:

$$\mathscr{I}(Y) = D_{\mathrm{KL}}\left(P_Y \parallel P_Y^P\right) = \int P_Y(y) \log \frac{P_Y(y)}{\prod_i P_{Y_i}(y_i)}\, dy. \tag{2.5}$$

A Pythagorean identity exists on the Product manifold. For any candidate product distribution $Q = \prod_i q_i$, the divergence splits precisely into the mutual information and the marginal divergences. Expanding the logarithm reveals this additive structure:

$$\begin{aligned} D_{\mathrm{KL}}\left(P_Y \| Q\right) &= \int P_Y(y) \log \left(\frac{P_Y(y)}{P_Y^P(y)} \frac{P_Y^P(y)}{Q(y)} \right) dy \\ &= \int P_Y(y) \log \frac{P_Y(y)}{P_Y^P(y)}\, dy + \sum_i \int P_{Y_i}(y_i) \log \frac{P_{Y_i}(y_i)}{q_i(y_i)}\, dy \\ &= \mathscr{I}(Y) + \sum_i D_{\mathrm{KL}}\left(P_{Y_i} \| q_i\right). \end{aligned} \tag{2.6}$$

A similar Pythagorean identity holds for the Gaussian manifold $\mathcal{G}$. Let $\mathcal{N}(\Sigma)$ denote an arbitrary zero-mean multivariate Gaussian distribution parameterized by a covariance matrix Σ. Furthermore, let $\mathcal{N}(\operatorname{Cov} Y)$ denote the specific Gaussian distribution that shares the covariance structure of the empirical data Y. Geometrically, $\mathcal{N}(\operatorname{Cov} Y)$ represents the orthogonal projection of P_Y onto the Gaussian manifold, yielding the best possible Gaussian approximation of the data.

To prove this decomposition, let $p(y)$ be the empirical density of P_Y, $\phi_{\Sigma_Y}(y)$ be the density of the best Gaussian fit $\mathcal{N}(\operatorname{Cov} Y)$, and $\phi_\Sigma(y)$ be the density of the arbitrary Gaussian target $\mathcal{N}(\Sigma)$. Expanding the Kullback-Leibler divergence integral yields:

$$\begin{aligned} D_{\mathrm{KL}}\big(P_Y \| \mathcal{N}(\Sigma)\big) &= \int p(y) \log \left(\frac{p(y)}{\phi_{\Sigma_Y}(y)} \frac{\phi_{\Sigma_Y}(y)}{\phi_\Sigma(y)} \right) dy \\ &= D_{\mathrm{KL}}\big(P_Y \| \mathcal{N}(\operatorname{Cov} Y)\big) + \int p(y) \log \frac{\phi_{\Sigma_Y}(y)}{\phi_\Sigma(y)} dy. \end{aligned} \tag{2.7}$$

Because the log-ratio of zero-mean Gaussians is strictly quadratic, its expected value depends entirely on the distribution's covariance matrix. Since P_Y and its projection $\mathcal{N}(\operatorname{Cov} Y)$ share the same covariance matrix by definition, we can swap their probability measures in the remaining integral without altering the result:

$$\int p(y) \log \frac{\phi_{\Sigma_Y}(y)}{\phi_\Sigma(y)} dy = \int \phi_{\Sigma_Y}(y) \log \frac{\phi_{\Sigma_Y}(y)}{\phi_\Sigma(y)} dy = D_{\mathrm{KL}}\big(\mathcal{N}(\operatorname{Cov} Y) \| \mathcal{N}(\Sigma)\big). \tag{2.8}$$

Substituting this identity back into the expansion yields the final Pythagorean theorem. The divergence to the arbitrary target decomposes into the divergence to the projection and the divergence between the Gaussians:

$$D_{\mathrm{KL}}\big(P_Y \| \mathcal{N}(\Sigma)\big) = D_{\mathrm{KL}}\big(P_Y \| \mathcal{N}(\operatorname{Cov} Y)\big) + D_{\mathrm{KL}}\big(\mathcal{N}(\operatorname{Cov} Y) \| \mathcal{N}(\Sigma)\big). \tag{2.9}$$

2.2.3 Independence, Correlation, and Non-Gaussianity

These geometric identities can be combined to relate independence directly to non-Gaussianity. We define the non-Gaussianity of the joint variable Y as $G(Y) = D_{\mathrm{KL}}(P_Y \,\|\, \mathcal{N}(\operatorname{Cov} Y))$, and the scalar measure of correlation as $C(Y) = D_{\mathrm{KL}}(\mathcal{N}(\operatorname{Cov} Y) \,\|\, \mathcal{N}(\operatorname{diag} \operatorname{Cov} Y))$.

To establish the relation between these quantities, we evaluate the KL divergence from the empirical distribution P_Y to a fully factorized Gaussian target $Q = \mathcal{N}(\operatorname{diag} \operatorname{Cov} Y)$. We can compute this divergence via two distinct geometric paths shown in Figure 2.1.

Following the Pythagorean identity on the Product manifold (Equation 2.6):

$$D_{\mathrm{KL}}\big(P_Y \| \mathcal{N}(\operatorname{diag}\operatorname{Cov} Y)\big) = \mathscr{I}(Y) + \sum_i G(Y_i). \tag{2.10}$$

Following the Pythagorean identity on the Gaussian manifold (Equation 2.9):

$$D_{\mathrm{KL}}\big(P_Y \| \mathcal{N}(\operatorname{diag}\operatorname{Cov} Y)\big) = G(Y) + C(Y). \tag{2.11}$$

Equating these two paths yields Cardoso's unified theorem (Cardoso, 2022):

$$\mathscr{I}(Y) + \sum_i G(Y_i) = G(Y) + C(Y). \tag{2.12}$$

To demonstrate how maximizing marginal non-Gaussianities minimizes mutual information, we analyze the terms under the constraints of the ICA pipeline. Crucially, the joint non-Gaussianity $G(Y)$ is invariant under any invertible linear transformation (Cardoso, 2022). Furthermore, once the data has been whitened, the covariance matrix is forced to be the identity matrix. This implies that $\operatorname{Cov}(Y) = \operatorname{diag}\operatorname{Cov}(Y)$, which strictly forces the correlation term to zero ($C(Y) = 0$).

Applying these constraints, the relationship rearranges into a direct subtraction:

$$\begin{aligned} \mathscr{I}(Y) + \sum_i G(Y_i) &= G(Y) + 0 \\ \mathscr{I}(Y) &= G(Y) - \sum_i G(Y_i). \end{aligned} \tag{2.13}$$

Because $G(Y)$ remains constant under orthogonal rotation, minimizing the mutual information $\mathscr{I}(Y)$, finding independent components, is mathematically equivalent to maximizing the sum of the marginal non-Gaussianities $\sum_i G(Y_i)$. This shows that centering and whitening constrain the data to a subspace where statistical independence can be achieved solely by maximizing non-Gaussianity.

2.3 Optimal Transport Theory

As established in the previous sections, the core objective of ICA is to identify independent components by maximizing the non-Gaussianity of linear projections. While classical methods rely on surrogate approximations to evaluate this divergence from a Gaussian state, we propose adopting a true metric on the space of probability distributions. Optimal transport theory provides this metric, which evaluates the distance of probability measures. By replacing local proxy contrasts with this true metric, we can quantify non-Gaussianity as the optimal transportation cost between a projected empirical distribution and a standard Gaussian target.

2.3.1 The Monge Problem: Push-Forwards and Pull-Backs

While information geometry utilizes Kullback-Leibler divergence, optimal transport provides an alternative metric space parameterized by the physical geometry of probability mass. First formulated by Gaspard Monge (1781), the problem was originally conceptualized as finding the most cost-effective way to move a distribution of mass.

Mathematically, this is framed as finding a transport map T that associates points from a continuous source measure μ to a target measure ν. We express this spatial modification of probability mass using the *push-forward* operator $T_\#$. A measure ν is the push-forward of μ by a map T, denoted $\nu = T_\#\mu$, if for any measurable set B in the target space:

$$\nu(B) = \mu(T^{-1}(B)). \tag{2.14}$$

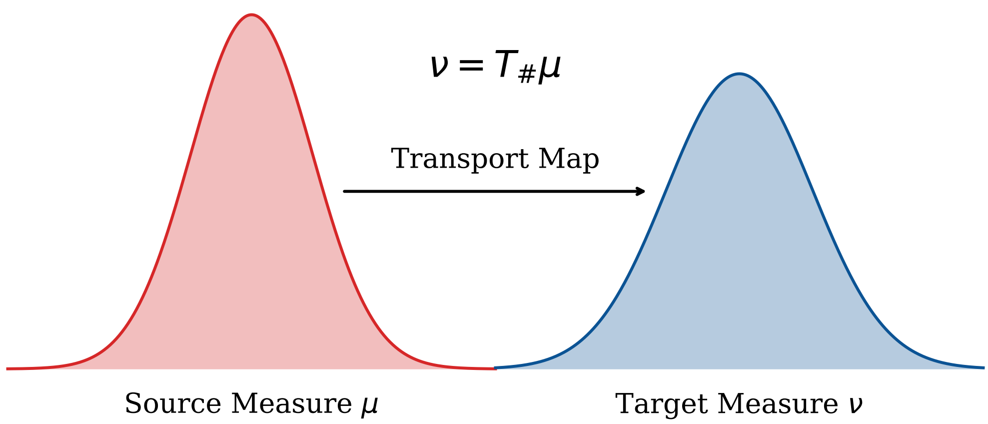

Figure 2.2: *The Monge transport map T pushes forward the probability mass from the source measure μ to construct the target measure ν, strictly conserving total mass during the transformation.*

As illustrated in Figure 2.2, the push-forward operator acts on measures, effectively pushing mass forward along the map T. It is conceptually distinct from the *pull-back* operator $T^\#$, which acts on functions. For a continuous, bounded test function $h : \mathcal{Y} \to \mathbb{R}$, the pull-back evaluates the function via the composition mapping ($T^\# h = h \circ T$). Because push-forward and pull-back are adjoint operators, the Monge problem seeks to minimize a transportation cost function $c(x, y)$ subject to the strict constraint that $T_{\#}\mu = \nu$:

$$\min_T \left\{ \int_{\mathcal{X}} c(x, T(x)) d\mu(x) \quad : \quad T_{\#}\mu = \nu \right\}. \tag{2.15}$$

2.3.2 The Kantorovich Relaxation

The fundamental limitation of the Monge problem is that the map T must be strictly deterministic; mass from a single source location cannot be split to multiple targets. Consequently, if the target distribution has more discrete points than the source distribution, a valid Monge map may not even exist.

To resolve this degeneracy, Kantorovich (1942) relaxed the deterministic requirement, proposing instead a probabilistic transport that permits mass splitting. The Kantorovich formulation replaces the deterministic map T with a joint probability distribution, or coupling, $\pi \in \Pi(\mu, \nu)$, which dictates how fractions of mass are distributed from the source to the target.

Definition 2.1 The Kantorovich Coupling Space. *Let $\mathcal{X}$ and $\mathcal{Y}$ be the source and target measurable spaces. The space of valid couplings $\Pi(\mu, \nu)$ is formally defined as the collection of all joint probability measures π on the product space $\mathcal{X} \times \mathcal{Y}$ whose marginal distributions are exactly μ and ν. Mathematically, for any measurable subsets $A \subset \mathcal{X}$ and $B \subset \mathcal{Y}$:*

$$\Pi(\mu, \nu) = \big\{ \pi \in \mathcal{P}(\mathcal{X} \times \mathcal{Y}) \mid \pi(A \times \mathcal{Y}) = \mu(A), \text{ and } \pi(\mathcal{X} \times B) = \nu(B) \big\}. \tag{2.16}$$

As illustrated in Figure 2.3, this relaxation is universally applicable across three fundamental statistical scenarios. In the discrete setting depicted in the first column, the coupling manifests as a transportation matrix mapping distinct point masses. In the semi-discrete setting, mass from a continuous density is optimally partitioned into specific discrete target regions. In the fully continuous setting, the coupling forms a joint density over the continuous supports. By allowing fractional transport in all domains, Kantorovich transformed the combinatorial Monge problem into a more tractable optimization problem.

2.3.3 The Wasserstein Distances (W_1 and W_2)

The minimal cost required to transform one probability distribution into another is formalized as the Wasserstein distance (Villani, 2003). For continuous measures in arbitrary dimensions d, utilizing a Euclidean distance cost function $c(x, y) = \|x - y\|^p$, the general W_p distance is defined over the space of optimal couplings

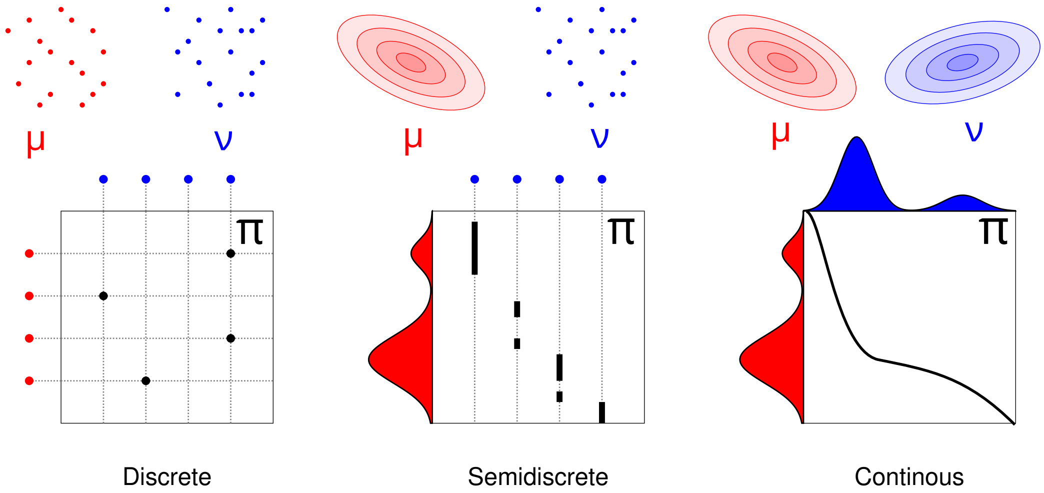

Figure 2.3: *A schematic view of the input measures and optimal couplings across the three primary scenarios of Kantorovich Optimal Transport: Discrete, Semi-discrete, and Continuous. The top row depicts the marginal distributions, while the bottom row visualizes the structural nature of the corresponding coupling spaces. Figure adapted from Peyré and Cuturi (2019).*

$\Pi(\mu, \nu)$ as:

$$W_p(\mu, \nu) = \left(\inf_{\pi \in \Pi(\mu,\nu)} \int_{\mathbb{R}^d \times \mathbb{R}^d} \|x - y\|^p \, d\pi(x, y) \right)^{1/p}. \tag{2.17}$$

While the integral definition above correctly formalizes optimal transport in high dimensions, direct analytical computation of $\pi(x, y)$ is highly restrictive for arbitrary multivariable spaces. The concept of sorting data to construct a cumulative distribution function (CDF) is strictly a one-dimensional phenomenon, as there is no natural, universal ordering of coordinates in $\mathbb{R}^d$ $(d > 1)$.

However, in the specific one-dimensional setting, the geometry of optimal transport drastically simplifies (Villani, 2003, Chapter 2). Because the data can be sorted, the W_1 distance simplifies to the integral of the absolute difference between the univariate cumulative distribution functions F_μ and F_ν:

$$W_1(\mu, \nu) = \int_{-\infty}^{\infty} |F_\mu(x) - F_\nu(x)| \, dx. \tag{2.18}$$

Similarly, the squared Wasserstein distance $(p = 2)$, which penalizes local geometric distortions, takes on a highly efficient analytical form in one dimension. It is most conveniently expressed using the quantile functions, which are the inverses of the cumulative distributions (F_μ^{-1} and F_ν^{-1}):

$$W_2^2(\mu, \nu) = \int_0^1 |F_\mu^{-1}(t) - F_\nu^{-1}(t)|^2 \, dt. \tag{2.19}$$

By avoiding the need to construct multi-dimensional couplings, the 1D W_2^2 metric provides a computationally efficient mechanism for measuring structural divergence. Crucially, unlike the local proxy approximations utilized in classical ICA algorithms, the Wasserstein distance is a true metric on the space of probability distributions (Villani, 2003, Chapter 7). Therefore, by evaluating linear projections against standard normal, the 1D squared Wasserstein distance emerges as a reliable contrast function for our proposed optimal transport ICA framework.

Chapter 3

The Optimal Transport ICA (OT-ICA) Framework

3.1 Optimal Transport Distance as a Contrast for ICA

As established in the preceding chapter, the process of centering and whitening restricts the search for independent components to the manifold of orthogonal matrices. Furthermore, Cardoso's unified theorem (Cardoso, 2022) demonstrates that within this whitened space, the mutual information of the projections is minimized precisely when the sum of their marginal non-Gaussianities is maximized. Traditional Independent Component Analysis algorithms approach this optimization by utilizing proxy contrast functions, such as kurtosis or negentropy approximations. While computationally inexpensive, these proxies do not evaluate the full probability density.

This thesis formulates ICA directly as an optimal transport problem, utilizing the squared Wasserstein distance, W_2^2, as a direct measure of non-Gaussianity. We formalize this framework as follows:

Definition 3.1 The OT-ICA Optimization Objective. *Let* $\mathbf{X}$ *be the whitened observed mixture, and let* $\mathbf{b} \in S_n$ *be a candidate projection vector on the unit sphere. Let the empirical distribution of the projected data be denoted by* $\mathbb{P}_{\mathbf{b}^\top \mathbf{X}}$*, and let the standard normal distribution be defined as* $\Gamma \equiv \mathcal{N}(0,1)$*. The Optimal Transport ICA (OT-ICA) objective is to find the orthogonal rotation matrix whose column vectors* $\mathbf{b}$ *maximize the distance to* Γ *(maximize non-gaussianity):*

$$\hat{\mathbf{b}} = \arg\max_{\mathbf{b} \in S_n} W_2^2\big(\mathbb{P}_{\mathbf{b}^\top \mathbf{X}}, \Gamma\big). \tag{3.1}$$

By framing the problem in this manner, we leverage the optimal transport metric's ability to evaluate geometric deviations from Gaussianity, providing a sensitive contrast function.

3.2 Theoretical Motivation: Bounding Mixture Non-Gaussianity

To justify the use of W_2^2 as an objective function for ICA, we demonstrate that maximizing this distance recovers the true independent sources. By the Central Limit Theorem, a mixture of non-Gaussian sources is more Gaussian than the constituent sources themselves. In optimal transport terms, this implies that the Wasserstein distance between a mixture and a Gaussian distribution is bounded by the distances of its independent components.

3.2.1 The W_2 Upper Bound

Theorem 3.1 **W_2 Upper Bound.** *Let $\mathbf{Z} = (Z_1, \ldots, Z_d)^\top$ be a vector of mutually independent latent sources, and let $\mathbf{b}$ be a weight vector on the unit sphere ($\sum_{i=1}^d b_i^2 = 1$). The squared Wasserstein distance between the linear mixture $\mathbf{b} \cdot \mathbf{Z}$ and the standard normal distribution Γ is bounded by the weighted sum of the source distances to Γ:*

$$W_2^2(\mathbf{b} \cdot \mathbf{Z}, \Gamma) \leq \sum_{i=1}^{d} b_i^2 W_2^2(Z_i, \Gamma). \tag{3.2}$$

Proof. We construct a specific joint probability distribution, or coupling, using a common source of randomness. Let $\mathbf{N} = (N_1, \ldots, N_d)^\top$ be a vector of d independent standard normal variables, $\mathbf{N} \sim \mathcal{N}(0, \mathbf{I}_d)$, assumed to be statistically independent of the sources $\mathbf{Z}$. For each independent latent source Z_i, there exists an optimal transport map T_i that pushes forward the standard normal distribution to the source distribution, denoted as $(T_i)_\#\Gamma = Z_i$.

We construct a random variable X representing the mixture using the transport maps evaluated on the common noise vector:

$$X = \sum_{i=1}^{d} b_i T_i(N_i). \tag{3.3}$$

Because $T_i(N_i)$ follows the distribution of Z_i, X models the weighted mixture $\mathbf{b} \cdot \mathbf{Z}$. To compare this mixture to a Gaussian distribution, we construct a target variable

Y using the same underlying noise vector $\mathbf{N}$ and weights $\mathbf{b}$:

$$Y = \sum_{i=1}^{d} b_i N_i. \tag{3.4}$$

Given that the weights sum to unity and the components N_i are independent standard normals, the resulting variable Y follows the standard normal distribution, $Y \sim \Gamma$. The pair (X, Y) creates a valid coupling between the mixture distribution and the Gaussian target.

The squared Wasserstein distance is the infimum of the expected squared cost over all valid couplings. Therefore, the optimal distance is less than or equal to the cost of our constructed coupling (X, Y), yielding the inequality:

$$W_2^2(\mathbf{b} \cdot \mathbf{Z}, \Gamma) \leq \mathbb{E}\left[|X - Y|^2\right]. \tag{3.5}$$

Expanding the squared difference yields:

$$\begin{aligned} |X - Y|^2 &= \left| \sum_{i=1}^{d} b_i T_i(N_i) - \sum_{i=1}^{d} b_i N_i \right|^2 \\ &= \left(\sum_{i=1}^{d} b_i \big(T_i(N_i) - N_i\big) \right)^2. \end{aligned} \tag{3.6}$$

Squaring the summation produces diagonal terms ($i = j$) and cross terms ($i \neq j$):

$$|X - Y|^2 = \sum_{i=1}^{d} b_i^2 \big(T_i(N_i) - N_i\big)^2 + \sum_{i \neq j} b_i b_j \big(T_i(N_i) - N_i\big)\big(T_j(N_j) - N_j\big). \tag{3.7}$$

Because the underlying noise variables N_i and N_j are independent, and the functions evaluate to a mean of zero, when we take the expectation, all cross terms vanish. We are left with the expectation of the diagonal terms:

$$\mathbb{E}\left[|X - Y|^2\right] = \sum_{i=1}^{d} b_i^2 \mathbb{E}\left[\big(T_i(N_i) - N_i\big)^2\right]. \tag{3.8}$$

Recognizing that $\mathbb{E}[(T_i(N_i) - N_i)^2]$ is the definition of the squared Wasserstein distance between the individual source Z_i and Γ, we arrive at the bound:

$$W_2^2(\mathbf{b} \cdot \mathbf{Z}, \Gamma) \leq \sum_{i=1}^{d} b_i^2 W_2^2(Z_i, \Gamma). \quad \blacksquare \tag{3.9}$$

By constructing a valid joint coupling, we establish that cross-dimensional terms vanish under expectation, proving that mixture non-Gaussianity is bounded by the weighted sum of the constituent sources. This property acts as an optimal transport analogue to the Central Limit Theorem, demonstrating that linear mixing reduces the overall distance to Gaussianity. Consequently, maximizing this distance forces the weight vector $\mathbf{b}$ to collapse toward a canonical axis, effectively isolating a single independent component.

3.2.2 Comonotonicity and the Proof of Strict Inequality

Optimizing ICA requires establishing that mixtures are strictly less non-Gaussian than the original sources. We determine the conditions under which the upper bound becomes a strict inequality.

Theorem 3.2 Strict W_2 Inequality for Non-Gaussian Sources. *Assuming at most one source Z_i is Gaussian, and the source distributions possess smooth and strictly positive densities, the upper bound derived in Theorem 3.1 is a strict inequality for any valid mixture where multiple weights b_i are non-zero:*

$$W_2^2(\mathbf{b} \cdot \mathbf{Z}, \Gamma) < \sum_{i=1}^{d} b_i^2 W_2^2(Z_i, \Gamma). \tag{3.10}$$

Proof. In a one-dimensional setting, the W_2 distance equals the expected squared difference of a specific coupling if and only if the random variables are comonotonic. By Brenier's Theorem (Brenier, 1991), this implies there exists a strictly increasing, deterministic function mapping one variable to the other. For the constructed mixture variable X and the Gaussian target Y to be comonotonic with respect to the shared noise vector $\mathbf{N} \in \mathbb{R}^d$, their gradients must be parallel at every point in the probability space. This requires the existence of a scalar function $\lambda(\mathbf{N})$ such that:

$$\nabla X(\mathbf{N}) = \lambda(\mathbf{N}) \nabla Y(\mathbf{N}). \tag{3.11}$$

Assuming the source distributions possess smooth and strictly positive densities, Caffarelli's regularity theory (Caffarelli, 1992) guarantees that the optimal transport maps T_i are continuously differentiable. We compute the gradients of our constructed variables with respect to $\mathbf{N}$. The gradient of the Gaussian target is the weight vector:

$$Y(\mathbf{N}) = \sum_{i=1}^{d} b_i N_i \implies \nabla Y = \mathbf{b}. \tag{3.12}$$

The gradient of the source mixture leverages the derivatives of the individual transport maps:

$$X(\mathbf{N}) = \sum_{i=1}^{d} b_i T_i(N_i) \implies \nabla X = \begin{bmatrix} b_1 T_1'(N_1) \\ b_2 T_2'(N_2) \\ \vdots \\ b_d T_d'(N_d) \end{bmatrix}. \tag{3.13}$$

Substituting these gradients into the comonotonicity condition yields the system of equations:

$$\nabla X = \lambda(\mathbf{N})\mathbf{b}. \tag{3.14}$$

To determine if this condition holds, we differentiate both the Left-Hand Side (LHS) and the Right-Hand Side (RHS) with respect to $\mathbf{N}$ to compute their Hessian matrices.

On the LHS, because X separates into independent terms $b_i T_i(N_i)$, the cross-derivatives $\frac{\partial^2 X}{\partial N_i \partial N_j}$ are zero for all $i \neq j$. This yields a diagonal matrix containing the second derivatives of the transport maps. On the RHS, we differentiate the product $\lambda(\mathbf{N})\mathbf{b}$. Applying the product rule yields the outer product of $\mathbf{b}$ and $\nabla\lambda$, forming a Rank-1 matrix. Equating the matrices:

$$\underbrace{\begin{bmatrix} b_1 T_1''(N_1) & 0 & \cdots & 0 \\ 0 & b_2 T_2''(N_2) & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & b_d T_d''(N_d) \end{bmatrix}}_{\text{Hessian of } X \text{ (Diagonal)}} = \underbrace{\begin{bmatrix} b_1 \frac{\partial\lambda}{\partial N_1} & b_1 \frac{\partial\lambda}{\partial N_2} & \cdots & b_1 \frac{\partial\lambda}{\partial N_d} \\ b_2 \frac{\partial\lambda}{\partial N_1} & b_2 \frac{\partial\lambda}{\partial N_2} & \cdots & b_2 \frac{\partial\lambda}{\partial N_d} \\ \vdots & \vdots & \ddots & \vdots \\ b_d \frac{\partial\lambda}{\partial N_1} & b_d \frac{\partial\lambda}{\partial N_2} & \cdots & b_d \frac{\partial\lambda}{\partial N_d} \end{bmatrix}}_{\text{Outer Product } \mathbf{b}(\nabla\lambda)^\top \text{ (Rank-1)}} \tag{3.15}$$

Examining any off-diagonal entry located at row i and column j ($i \neq j$), the diagonal LHS matrix dictates that this entry must be zero. The RHS Rank-1 matrix evaluates this entry as $b_i \frac{\partial\lambda}{\partial N_j}$, establishing the requirement:

$$b_i \frac{\partial\lambda}{\partial N_j} = 0 \quad \text{for all } i \neq j. \tag{3.16}$$

Because we assume a mixture where multiple component weights are non-zero ($b_i \neq 0$), it follows that $\frac{\partial\lambda}{\partial N_j} = 0$. Since this holds across all dimensions, the gradient of the scaling function is zero ($\nabla\lambda = \mathbf{0}$). Consequently, $\lambda(\mathbf{N})$ is a global scalar constant, λ.

Equating the diagonal elements implies $T_i'(N_i) = \lambda$. Integrating this relationship indicates the optimal transport maps are linear functions of the form:

$$T_i(x) = \lambda x + c. \tag{3.17}$$

Applying a purely linear transformation to Gaussian noise N_i produces another Gaussian distribution. The identifiability assumption of ICA dictates that the original latent sources Z_i are non-Gaussian, which requires their optimal transport mappings from a Gaussian base to be non-linear. Therefore, the linear map requirement contradicts the non-Gaussianity of the sources. Because the condition for equality cannot be met, the relationship is a strict inequality.

$$W_2^2(\mathbf{b} \cdot \mathbf{Z}, \Gamma) < \sum_{i=1}^{d} b_i^2 W_2^2(Z_i, \Gamma). \quad \blacksquare \tag{3.18}$$

The practical implication of this strict inequality dictates the behavior of the objective function. The upper bound is structured as a convex combination of the marginal non-Gaussianities, subject to the constraint $\sum_{i=1}^{d} b_i^2 = 1$. When the objective function is maximized, the supremum is achieved when the entirety of the weight is assigned to the single latent source exhibiting the maximum distance to Gaussianity. Consequently, finding the global maximum forces the weight vector $\mathbf{b}$ to collapse to a canonical axis ($b_k^2 = 1$ and $b_{i\neq k}^2 = 0$), thereby isolating the most non-Gaussian independent component from the mixture.

3.2.3 Sequential Extraction via Orthogonal Deflation

Maximizing the W_2^2 objective isolates only the most non-Gaussian source, but the framework requires a mechanism to unmix the remaining $(d-1)$ latent components. This is achieved through a sequential deflation process.

Let $\mathbf{b}_1$ be the canonical weight vector recovered by the initial maximization. To extract the second component, the optimization must be restricted to the orthogonal complement of $\mathbf{b}_1$. We update the search space by constraining the subsequent weight vector $\mathbf{b}_2$ to be strictly orthogonal to the first ($\mathbf{b}_2^\top \mathbf{b}_1 = 0$).

For the k-th component, the maximization is performed under the constraint that the new weight vector $\mathbf{b}_k$ satisfies $\mathbf{b}_k^\top \mathbf{b}_j = 0$ for all previously extracted components $j < k$. By iteratively extracting the maximum of the objective function and restricting the subsequent search to the remaining orthogonal subspace, the algorithm sequentially constructs the full orthogonal unmixing matrix $\mathbf{B} \in \mathbb{R}^{d\times d}$. This guarantees that all d mutually independent sources are recovered.

3.3 Empirical Landscape: W_1 versus W_2^2

Having established the theoretical foundations of the W_2^2 metric, we evaluate its practical suitability for manifold optimization. In the context of ICA, gradient ascent algorithms must climb the loss landscape to locate the maxima of the distance metric, thereby isolating the independent sources.

While the W_1 distance is robust to outliers, its reliance on an L_1 absolute penalty introduces geometric challenges when applied to finite empirical samples. We demonstrate why the squared L_2 penalty of the W_2^2 metric is preferable for gradient-based convergence.

3.3.1 Empirical Gradients and Derivative Volatility

To observe this behavior, we generated a synthetic bivariate mixture of Student-t distributions ($\nu = 4$) and evaluated both the raw distances and their first-order numerical gradients across a continuous sweep of rotation angles $\theta \in [0, \pi]$. Because the unmixing matrix restricts the search to orthogonal vectors, the two true independent components are geometrically separated by $\pi/2$ radians (90°).

The empirical results, visualized in Figure 3.1, illustrate the behaviors of the two metrics during continuous gradient ascent. In optimal transport, because the W_1 cost scales linearly with distance, the resulting optimization landscape lacks the convexity needed to smooth out finite sample noise.

The consequence of this lack of curvature is evident in the bottom gradient profile. Because W_1 relies on piecewise linear sorting mechanics without a squaring penalty, its empirical derivative fluctuates frequently as the projection angle rotates. Consequently, first-order gradient ascent solvers encounter a jagged derivative surface, which hinders optimizer convergence. This instability is exacerbated in higher dimensions, where the W_1 derivative surface becomes even more jagged and difficult to traverse. Furthermore, this volatility limits the utility of second-order Quasi-Newton approximations (such as L-BFGS), as a stable Hessian cannot be reliably estimated from such a jagged and difficult-to-traverse gradient.

In contrast, the squared penalty in the W_2^2 metric acts as a smoother, penalizing local geometric distortions. As seen in the gradient profiles, the W_2^2 derivative behaves continuously, decelerating predictably as it approaches the optimum. This retention of curvature provides a reliable signal for gradient-based solvers, allowing the OT-ICA framework to converge consistently.

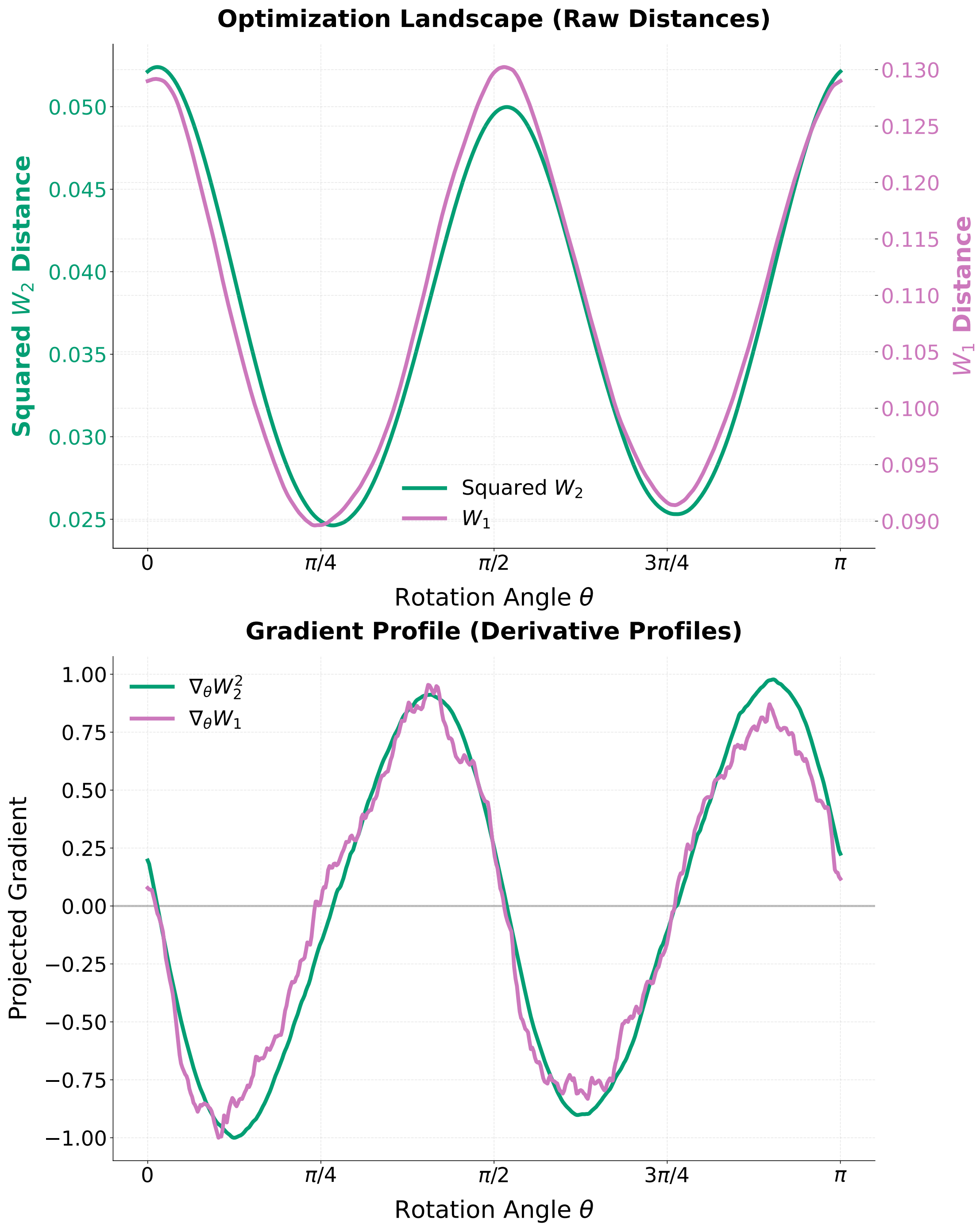

Figure 3.1: *The empirical optimization landscapes (top) and their normalized first derivatives (bottom). While both metrics peak near the true independent components ($\theta \approx 0, \pi/2, \pi$), their gradient behaviors differ. The W_2^2 gradient smoothly approaches zero, whereas the W_1 gradient exhibits jagged volatility on finite samples.*

Chapter 4

Algorithmic Enhancements

This chapter transitions the theoretical formulation of OT-ICA into a practical algorithm. While the baseline optimal transport formulation provides a valid objective, its reliance on discrete sorting and unconstrained gradient descent presents both algorithmic and computational bottlenecks. To resolve these issues, this chapter introduces a series of algorithmic enhancements. We develop analytical Gaussian targets to eliminate quantile approximation noise, formalize an optimization framework on the space of orthogonal matrices to strictly preserve component decorrelation, and utilize various methodologies to navigate discontinuous landscapes. The chapter concludes by combining these components into the complete OT-ICA algorithm.

4.1 The Baseline OT-ICA Algorithm

Having established the theoretical validity of the squared Wasserstein distance (W_2^2) for Independent Component Analysis, the challenge is to minimize this cost function computationally. Just like the standard FastICA algorithm (Hyvärinen et al., 2001), the baseline OT-ICA algorithm proceeds by first centering and whitening the observed mixture $\mathbf{X} \in \mathbb{R}^{d \times N}$, yielding the whitened data $\widetilde{\mathbf{X}}$. The algorithm initializes an unmixing matrix $\mathbf{B} \in \mathbb{R}^{d \times d}$ within the Orthogonal Group, denoted hereafter as the space $\mathcal{S}$, which is the mathematical set of all orthogonal matrices.

In the forward pass, the data is projected onto the candidate components, $\mathbf{Y} = \mathbf{B}\widetilde{\mathbf{X}}$. For each row of $\mathbf{Y}$, the empirical cumulative distribution function (CDF) is constructed by sorting the N observations. This sorting operation evaluates the inverse CDF, the quantile function, over a uniform probability grid, typically approximated as $p_i = (i - 0.5)/N$ (Villani, 2003, Chapter 2). The W_2^2 distance is computed by matching these sorted empirical quantiles to the corresponding quantiles of a standard normal distribution.

While theoretically sound, the standard implementation of this gradient descent loop faces geometric limitations. The optimal transport matching requires sorting at each iteration, presenting a computational bottleneck. Furthermore, standard gradient updates do not respect the strict orthogonality constraint of the mixing matrix, and discrete approximations of the target Gaussian introduce gradient noise. To make OT-ICA competitive with existing methods, specific algorithmic enhancements are required.

4.2 Computational Bottlenecks and Solutions

4.2.1 Analytical Gaussian Targets

One source of instability in the baseline algorithm is the discrete approximation of the target Gaussian distribution. Target quantiles could be evaluated at evenly spaced intervals ($q_i = \Phi^{-1}((i-0.5)/N)$). Representing a continuous Gaussian distribution with finite discrete points introduces approximation noise. During gradient computation, this noise translates into a non-smooth optimization landscape.

To eliminate this noise, we replace point-sampled targets with an analytical formulation. We compute the expected value of a standard normal variable within each discrete quantile bin.

Definition 4.1 Analytical Gaussian Target. *Let the uniform probability bin edges for N samples be defined as $p_i = \frac{i}{N}$ for $i = 0, \ldots, N$, with corresponding Gaussian domain boundaries $z_i = \Phi^{-1}(p_i)$, where Φ is the standard normal cumulative distribution function. The analytical target value T_i for the i-th sorted sample is the conditional expectation within that interval:*

$$T_i = N \int_{z_{i-1}}^{z_i} x\phi(x)\, dx \tag{4.1}$$

where $\phi(x)$ is the standard normal probability density function. Evaluating this integral yields the closed-form target:

$$T_i = N\big(\phi(z_{i-1}) - \phi(z_i)\big). \tag{4.2}$$

This discrete formulation computes the true expected Wasserstein cost only up to an additive constant. To demonstrate this, consider the squared Wasserstein distance between the sorted empirical projections y_i and the continuous Gaussian target $\Phi^{-1}(p)$ integrated over the uniform intervals $p \in [\frac{i-1}{N}, \frac{i}{N}]$:

$$W_2^2(P_N, \mathcal{N}(0,1)) = \sum_{i=1}^{N} \int_{\frac{i-1}{N}}^{\frac{i}{N}} \left(y_i - \Phi^{-1}(p)\right)^2 dp. \tag{4.3}$$

Expanding the squared binomial yields the empirical variance $\frac{1}{N}y_i^2$, the target Gaussian variance $\int \Phi^{-1}(p)^2 \, dp$, and the cross-term $-2y_i \int \Phi^{-1}(p) \, dp$. Because T_i is defined strictly as $N \int \Phi^{-1}(p) \, dp$, the discrete cost $\frac{1}{N} \sum (y_i - T_i)^2$ evaluates the same empirical variance and cross-terms as the continuous integral.

The two cost functions differ only in the final target variance term. Because the target Gaussian distribution is static, its variance is completely independent of the optimization matrix $\mathbf{B}$. Consequently, this difference is an absolute constant that vanishes under differentiation ($\nabla_{\mathbf{B}} C = \mathbf{0}$). By evaluating this analytical target once during initialization, the algorithm generates an accurate continuous Wasserstein gradient while maintaining the efficiency of discrete sorting.

4.2.2 Batched Vectorization for Restarts

Unlike Principal Component Analysis, which yields a unique global solution via singular value decomposition, the ICA optimization landscape is non-convex and characterized by local optima. Identifying the independent components requires multiple random initializations or restarts to explore the space.

Because evaluating the W_2^2 objective requires an $\mathcal{O}(N \log N)$ sorting operation for each candidate component, executing these random restarts sequentially is computationally expensive. We utilize a batched vectorization strategy. We aggregate N_{batch} random restarts into a single high-dimensional tensor $\mathbf{B}_{\text{batch}} \in \mathbb{R}^{N_{\text{batch}} \times d \times d}$.

This formulation allows the projection $\mathbf{Y} = \mathbf{B}_{\text{batch}} \widetilde{\mathbf{X}}$ to be computed simultaneously for all N_{batch} trajectories. The elements of the batch tensor are sorted in parallel. The pre-computed analytical target vector $\mathbf{T}$ is broadcast-subtracted across the batch to compute the distances and gradients in a single step.

4.2.3 The Riemannian Gradient

Because the input data is whitened, the unmixing matrix $\mathbf{B}$ must remain in the Orthogonal Group $\mathcal{S}$, satisfying $\mathbf{B}\mathbf{B}^\top = \mathbf{I}$. If we compute the standard Euclidean gradient of the squared Wasserstein distance, this vector points into the unconstrained ambient space $\mathbb{R}^{d \times d}$. Updating the matrix directly using this Euclidean gradient violates the strict orthogonality constraint.

To navigate the geometry of $\mathcal{S}$, we first compute the Euclidean gradient $\mathbf{G}$

of the empirical objective function in the ambient space. Let $\mathbf{Y} = \mathbf{B}\widetilde{\mathbf{X}}_{\text{batch}} \in \mathbb{R}^{d \times N_{\text{batch}}}$ be the projected mini-batch. For each projection component (row) i, there exists an optimal permutation π_i that sorts the empirical observations such that $y_{i,\pi_i(1)} \leq y_{i,\pi_i(2)} \leq \cdots \leq y_{i,\pi_i(N_{\text{batch}})}$. The total squared Wasserstein objective across all d dimensions is evaluated as:

$$W_2^2(\mathbf{B}) = \frac{1}{N_{\text{batch}}} \sum_{i=1}^{d} \sum_{j=1}^{N_{\text{batch}}} \left(y_{i,\pi_i(j)} - T_j\right)^2. \tag{4.4}$$

To compute the Euclidean gradient $\mathbf{G} = \nabla_{\mathbf{B}} W_2^2$, we differentiate this objective with respect to $\mathbf{B}$. Utilizing the envelope theorem for optimal transport (Santambrogio, 2015, Section 7.2), the variation of the optimal sorting map π_i evaluates to zero. This allows us to differentiate the squared cost while holding the permutations fixed. Taking the partial derivative with respect to a single row vector $\mathbf{b}_i$ yields:

$$\frac{\partial W_2^2}{\partial \mathbf{b}_i} = \frac{2}{N_{\text{batch}}} \sum_{j=1}^{N_{\text{batch}}} \left(y_{i,\pi_i(j)} - T_j\right) \widetilde{\mathbf{x}}_{\pi_i(j)}. \tag{4.5}$$

This expression evaluates the difference between the projection and the target, mapped back to the original data index. Let $\mathbf{T}^{\text{inv}} \in \mathbb{R}^{d \times N_{\text{batch}}}$ be the target matrix where the analytical targets T_j have been inversely permuted for each row to match the original, unsorted data ordering. The full Euclidean gradient expands into the closed-form matrix equation:

$$\mathbf{G} = \nabla_{\mathbf{B}} W_2^2 = \frac{2}{N_{\text{batch}}} \left(\mathbf{Y} - \mathbf{T}^{\text{inv}}\right) \widetilde{\mathbf{X}}_{\text{batch}}^{\top}. \tag{4.6}$$

To ensure the update step respects the orthogonality constraints, we project this Euclidean gradient $\mathbf{G}$ onto the tangent space of the manifold at the current evaluation point $\mathbf{B}$ as descibed in Definition 4.2. This geometric relationship is illustrated in Figure 4.1.

As depicted in the diagram, the current unmixing matrix $\mathbf{B}$ rests on the surface of the Orthogonal Matrices Space $\mathcal{S}$. The standard Euclidean gradient $\mathbf{G} = \nabla_{\mathbf{B}} W_2^2$ points outward into the unconstrained ambient space. To find a valid update direction, $\mathbf{G}$ is orthogonally projected onto the local tangent space $T_{\mathbf{B}}\mathcal{S}$. This projection yields the Riemannian gradient $\nabla_{\mathcal{S}}\mathbf{B}$, which represents the component of the gradient $\mathbf{G}$ that lies strictly along the geometric surface of $\mathcal{S}$ at point $\mathbf{B}$.

Definition 4.2 Riemannian Gradient on the Orthogonal Group. *Given the Euclidean gradient* $\mathbf{G}$ *and the current unmixing matrix* $\mathbf{B}$*, the Riemannian gradient* $\nabla_{\mathcal{S}}\mathbf{B}$ *resulting from the projection onto the tangent space* $T_{\mathbf{B}}\mathcal{S}$ *is given by*

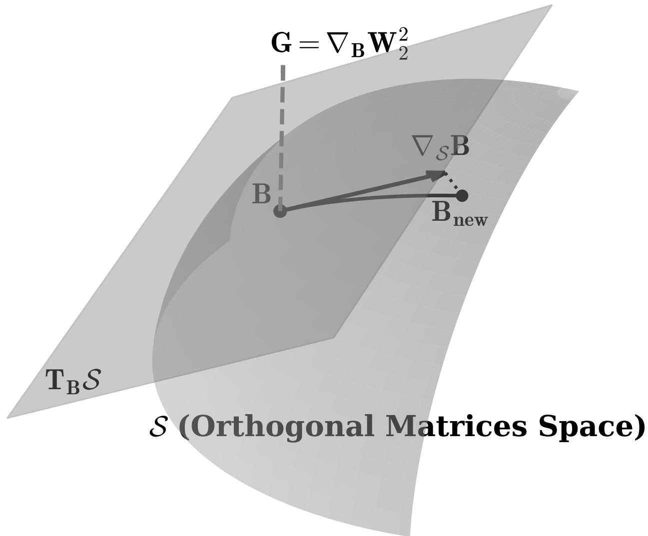

Figure 4.1: *Geometric sketch of optimization on the Orthogonal Group $\mathcal{S}$. The Euclidean gradient $\mathbf{G}$ is projected onto the tangent space $T_{\mathbf{B}}\mathcal{S}$ to yield the Riemannian gradient $\nabla_{\mathcal{S}}\mathbf{B}$. Following a step along the tangent plane, a retraction maps the estimate back to the orthogonal surface at $\mathbf{B}_{new}$.*

(Absil et al., 2008, Chapter 3):

$$\nabla_{\mathcal{S}}\mathbf{B} = \mathbf{G} - \frac{1}{2}\big(\mathbf{G}\mathbf{B}^\top + \mathbf{B}\mathbf{G}^\top\big)\mathbf{B}. \tag{4.7}$$

The term $(\mathbf{G}\mathbf{B}^\top + \mathbf{B}\mathbf{G}^\top)$ mathematically isolates the symmetric violation of orthogonality induced by the Euclidean gradient. Subtracting this term from $\mathbf{G}$ ensures the update step strictly aligns with the intrinsic tangent plane of the manifold, preserving the rotational structure of the matrix prior to retraction to $\mathbf{B}_{\text{new}}$.

Absil, Mahony, and Sepulchre (2008, Chapter 3 and 4) demonstrate that projecting the gradient onto the tangent plane prior to retraction is a numerical requirement. The normal component of the Euclidean gradient $\mathbf{G}$ points directly into the ambient space. Because this vector is orthogonal to the surface, it provides no optimization progress along the manifold. Furthermore, taking a direct step using $\mathbf{G}$ overshoots into the ambient space, degrading the local approximation bounds required to guarantee the convergence of the retraction mapping.

4.2.4 Retraction via Symmetric Decorrelation

Although the Riemannian gradient ensures the update step is tangential to the Orthogonal Group, moving along this tangent plane for a finite step size η causes the new matrix $\mathbf{B}_{\text{step}} = \mathbf{B} - \eta \nabla_{\mathcal{S}} \mathbf{B}$ to depart from the curved surface. To restore orthogonality, the matrix must be mapped back onto the manifold via a retraction (Absil et al., 2008, Chapter 4).

Definition 4.3 Symmetric Decorrelation Retraction. *Let $\mathbf{B}_{step}$ be the unmixing matrix after a tangent space update. To map the matrix back to the orthogonal surface, we first compute its overlap covariance matrix:*

$$\mathbf{C} = \mathbf{B}_{\text{step}} \mathbf{B}_{\text{step}}^{\top}. \tag{4.8}$$

If $\mathbf{B}_{step}$ were strictly orthogonal, $\mathbf{C}$ would equal the identity matrix $\mathbf{I}$. The retraction mapping corrects this deviation by applying the inverse square root of the covariance:

$$\mathbf{B}_{\text{new}} = \mathbf{C}^{-1/2} \mathbf{B}_{\text{step}}. \tag{4.9}$$

We can mathematically verify that this operation successfully retracts the matrix to the Orthogonal Group by evaluating the covariance of the updated matrix:

$$\mathbf{B}_{\text{new}} \mathbf{B}_{\text{new}}^{\top} = (\mathbf{C}^{-1/2} \mathbf{B}_{\text{step}})(\mathbf{B}_{\text{step}}^{\top} \mathbf{C}^{-1/2}) = \mathbf{C}^{-1/2} \mathbf{C} \mathbf{C}^{-1/2} = \mathbf{I}. \tag{4.10}$$

This retraction computes the orthogonal deviation ($\mathbf{C}$) and applies a uniform geometric correction ($\mathbf{C}^{-1/2}$). Because this correction is symmetric, it ensures that no single independent component is prioritized or artificially rotated during the retraction process, maintaining the stability of the batched optimization.

4.3 Total Complexity and Statistical Efficiency

We formalize the total computational cost of the OT-ICA framework. For a dataset of dimension d with N samples, K random restarts, and T optimization iterations, the per-iteration complexity is:

$$\mathcal{O}(K \cdot (d \cdot N \log N + d^3)) \tag{4.11}$$

The $N \log N$ term represents the sorting of projections, while the d^3 term represents the cost of the symmetric decorrelation used for retraction. For typical ICA applications where the number of samples exceeds the dimensionality ($N \gg d$), the d^3 cost is computationally negligible.

Beyond time complexity, we must account for statistical sample complexity. While the 1D projection maintains a convergence rate of $\mathcal{O}(N^{-1/2})$, the global identification of the mixing matrix is implicitly subject to the curse of dimensionality. Theoretical results in transport theory establish that the empirical Wasserstein distance in d dimensions converges at slower rates as dimension increases (Fournier & Guillin, 2015). As d expands, the number of samples N required to accurately resolve the maxima of non-Gaussianity on the manifold surface grows. An empirical evaluation of this statistical ceiling is reported in Chapter 5.

4.4 Optimization in Discrete Spaces

In the OT-ICA framework, applying continuous optimal transport directly to discrete mixtures introduces non-smooth regions that disrupt gradient calculation. In this section we introduce algorithmic enhancements to counter this difficulty.

4.4.1 Continuous Smoothing via Gaussian Dithering

We implement a Gaussian Dithering step to restore smooth gradients without resorting to $\mathcal{O}(N^2)$ entropic regularization techniques, such as Sinkhorn distances (Cuturi, 2013).

Adapted from signal processing techniques used to decorrelate quantization error (Schuchman, 1964), we inject a variance of continuous, zero-mean Gaussian noise ($\sigma^2_{\text{dither}} = 0.01$) into the 1D projected components $\mathbf{Y}$ prior to the sorting operation. The sorting operation acts as a strict discrete quantizer. When data clusters densely, ties create highly correlated, deterministic errors in the gradient. By injecting Gaussian dither before sorting, we decorrelate this deterministic quantization error, transforming it into independent noise that first-order gradient solvers can average out. Mathematically, this operation convolves the discrete empirical distribution with a Gaussian kernel (Figure 4.2).

The choice of $\sigma^2_{\text{dither}} = 0.01$ functions as a strictly bounded heuristic governed by the algorithmic pipeline. Because the input data is whitened, its covariance matrix is the identity matrix ($\mathbb{E}[\widetilde{\mathbf{x}}\widetilde{\mathbf{x}}^\top] = \mathbf{I}$). For any 1D projection component $\mathbf{y}_i = \mathbf{b}_i^\top \widetilde{\mathbf{x}}$, the variance is formulated as:

$$\mathrm{Var}(\mathbf{y}_i) = \mathbf{b}_i^\top \mathbb{E}[\widetilde{\mathbf{x}}\widetilde{\mathbf{x}}^\top]\mathbf{b}_i = \mathbf{b}_i^\top \mathbf{I}\mathbf{b}_i = \|\mathbf{b}_i\|^2. \tag{4.12}$$

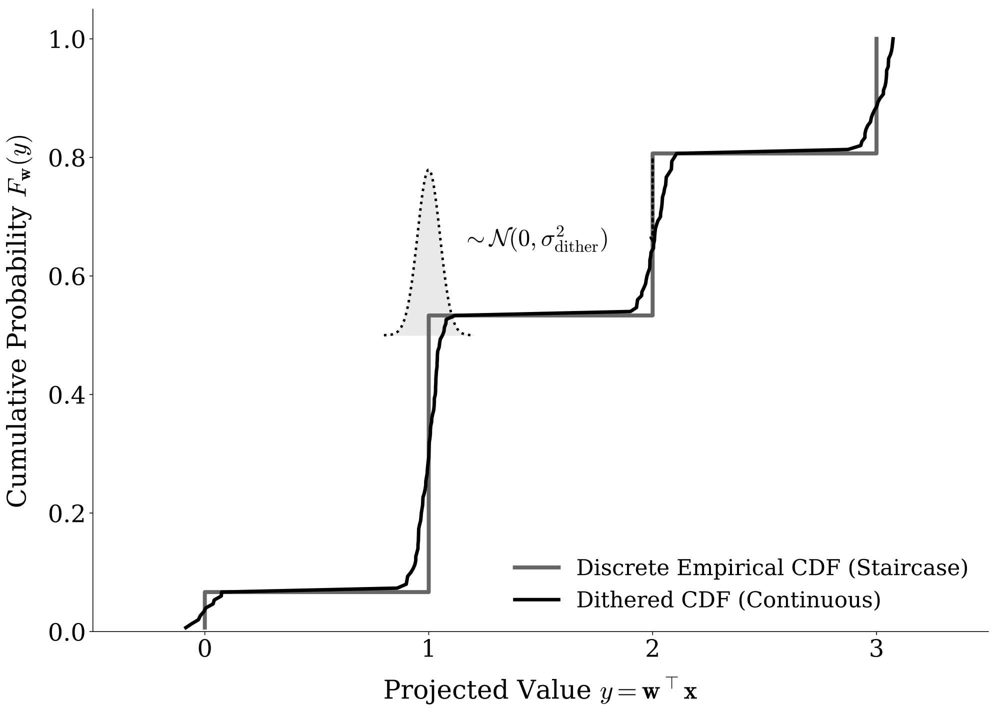

Figure 4.2: *The effect of algorithmic Gaussian Dithering. The non-differentiable staircase of a discrete empirical CDF (gray) is convolved with a narrow Gaussian kernel, yielding a strictly increasing, continuous curve (black).*

Furthermore, because the unmixing matrix is restricted to the Orthogonal Group, its row vectors must satisfy the strict orthogonality constraint:

$$\mathbf{B}\mathbf{B}^\top = \mathbf{I} \implies \|\mathbf{b}_i\|^2 = 1. \tag{4.13}$$

Therefore, the variance of any 1D projection within $\mathbf{Y}$ is locked to 1.0. Consequently, a dither variance of 0.01 corresponds to a standard deviation of 0.1, injecting noise scaled to 10% of the projection's standard deviation. This amplitude is sufficient to break deterministic sorting ties, yet small enough to preserve the geometric divergence from Gaussianity measured by the Wasserstein metric.

This deforms the step-wise discrete CDF into a continuous curve. Gaussian dithering restores stable gradients while preserving the $\mathcal{O}(N \log N)$ efficiency of the 1D optimal transport formulation. This acts as an uninformed regularizer applied to the projections $\mathbf{Y}$.

4.4.2 Escaping Local Minima with Stochastic Mini-Batching

Even with a smoothed CDF, the global optimization landscape of a discrete mixture remains non-convex. Evaluating the W_2 distance using the full dataset guarantees deterministic convergence to the nearest local maximum. In non-convex environments, this deterministic behavior prevents global convergence.

We utilize stochastic mini-batching. At each optimization step, the algorithm evaluates the Wasserstein gradients using a randomly sampled subset of the data (e.g., $N_{\text{batch}} = 512$). This stochastic subsampling introduces gradient noise. A geometric local minimum that exists for the full dataset presents a different gradient profile for a random subset. This stochastic momentum allows the optimizer to escape spurious local minima.

4.5 Limitations of Fixed-Point Algorithms for OT-ICA

A widely used algorithm for ICA, FastICA (Hyvärinen et al., 2001, Chapter 8), achieves efficient convergence by employing a second-order Newton fixed-point step. FastICA utilizes this approach by maximizing globally differentiable proxy contrast functions. A natural question is whether a similar fixed-point iteration can be derived for the OT-ICA objective (W_2^2).

Computing second-order Newton updates for the empirical Wasserstein metric is analytically intractable without introducing explicit density approximation mechanisms. Because the efficiency of 1D optimal transport relies on the empirical CDF generated via sorting, formulating the Hessian matrix $\mathbf{H} = \nabla_{\mathbf{w}}^2 W_2^2$ requires differentiating the transport map boundaries. This calculation strictly requires the continuous Probability Density Function (PDF) evaluated at the empirical quantiles. We provide the formal derivation showing this intractability in Appendix A.2.

Satisfying this requirement necessitates density approximation methods, such as Kernel Density Estimation (KDE). These methods introduce statistical noise and additional computational overhead, eating into the efficiency of the sorting-based transport formulation.

Consequently, the chosen strategy for the OT-ICA framework in this work utilizes Quasi-Newton methods, specifically the Limited-memory BFGS (L-BFGS) algorithm (Liu & Nocedal, 1989) adapted for the Orthogonal Group (Absil et al., 2008, Chapter 6). L-BFGS approximates the inverse Hessian curvature strictly by observing the history of first-order Wasserstein gradients, achieving convergence while preserving the efficiency of the sorting-based metric.

4.6 OT-ICA Algorithm Overview

The theoretical components and algorithmic enhancements are synthesized into the final OT-ICA procedure. The pseudo-code for extracting independent components using stochastic Riemannian optimization is detailed below (Algorithm 1).

Algorithm 1 The Optimal Transport ICA (OT-ICA) Algorithm

Require: Observed mixture $\mathbf{X} \in \mathbb{R}^{d \times N}$, iterations K, learning rate η, batch size N_{batch}, dither noise σ_{dither}

Ensure: Estimated unmixing matrix $\mathbf{B}_{\text{final}} \in \mathbb{R}^{d \times d}$

Phase 1: Preprocessing & Target Generation

Center the data: $\mathbf{X} \leftarrow \mathbf{X} - \mathbb{E}[\mathbf{X}]$

Whiten the data: $\widetilde{\mathbf{X}} \leftarrow (\mathbf{X}\mathbf{X}^\top)^{-1/2}\mathbf{X}$ ▷ Yields unit variance, uncorrelated data

Compute analytical target $\mathbf{T} \in \mathbb{R}^{N_{\text{batch}}}$ via analytical Gaussian CDF integration

Phase 2: Initialization

Initialize $\mathbf{B} \in \mathbb{R}^{d \times d}$ randomly from $\mathcal{N}(0, 1)$

Apply symmetric decorrelation:

$\mathbf{B} \leftarrow (\mathbf{B}\mathbf{B}^\top)^{-1/2}\mathbf{B}$ ▷ Initialize in the Orthogonal Group $\mathcal{S}$

Phase 3: Stochastic Riemannian Optimization

for $k = 1$ to K **do**

Sample stochastic mini-batch $\widetilde{\mathbf{X}}_{\text{batch}} \in \mathbb{R}^{d \times N_{\text{batch}}}$ from $\widetilde{\mathbf{X}}$

Project data onto candidates: $\mathbf{Y} \leftarrow \mathbf{B}\widetilde{\mathbf{X}}_{\text{batch}}$

Apply Gaussian Dithering: $\mathbf{Y}_{\text{dither}} \leftarrow \mathbf{Y} + \mathcal{N}(0, \sigma^2_{\text{dither}})$

Sort each row of $\mathbf{Y}_{\text{dither}}$ ascending to yield empirical quantiles $\mathbf{Y}_{\text{sorted}}$

Compute Wasserstein objective:

$\mathcal{L} = \frac{1}{N_{\text{batch}}} \sum_{i=1}^{d} ||\mathbf{Y}^{(i)}_{\text{sorted}} - \mathbf{T}||_2^2$

Backpropagate to compute Euclidean gradient: $\mathbf{G} = \nabla_{\mathbf{B}}\mathcal{L}$

Project to tangent space (Riemannian Gradient):

$\nabla_{\mathcal{S}}\mathbf{B} = \mathbf{G} - \frac{1}{2}\big(\mathbf{G}\mathbf{B}^\top + \mathbf{B}\mathbf{G}^\top\big)\mathbf{B}$

Update matrix along tangent plane:

$\mathbf{B}_{\text{step}} = \mathbf{B} + \eta\nabla_{\mathcal{S}}\mathbf{B}$ ▷ Gradient Ascent

Retract to Orthogonal Group $\mathcal{S}$ via symmetric decorrelation:

$\mathbf{B} \leftarrow (\mathbf{B}_{\text{step}}\mathbf{B}_{\text{step}}^\top)^{-1/2}\mathbf{B}_{\text{step}}$

Optional: Apply learning rate decay $\eta \leftarrow \eta \cdot 0.99$

end for

Phase 4: Finalization

$\mathbf{B}_{\text{final}} \leftarrow \mathbf{B}(\mathbf{X}\mathbf{X}^\top)^{-1/2}$ ▷ Map back to original data space

return $\mathbf{B}_{\text{final}}$

Chapter 5

Experimental Evaluation and Applications

This chapter evaluates the OT-ICA framework through empirical experiments and interdisciplinary applications. We first validate the baseline optimal transport methodology before evaluating its robustness on generalized hybrid mixtures. We then analyze the limitations of FastICA using engineered non-Gaussian distributions, followed by an investigation into the optimization bottlenecks presented by discrete data. The chapter proceeds to quantify the computational scaling trade-offs of the OT-ICA framework and concludes by applying it to clinical electroencephalography (EEG) artifact removal and econometric price discovery applications.

5.1 Evaluation Metrics and Methodology

5.1.1 The Amari Performance Index and Generalized Permutation Matrix

To empirically evaluate the success of an Independent Component Analysis (ICA) algorithm, a quantitative metric is required to measure the distance between the estimated unmixing matrix $\mathbf{B}_{\text{est}}$ and the true mixing matrix $\mathbf{A}$. Because ICA is subject to inherent scaling and permutation ambiguities, the target recovery is not the identity matrix, but rather a generalized permutation matrix.

Definition 5.1 Generalized Permutation Matrix. *A generalized permutation matrix is a square matrix containing exactly one non-zero entry in each row and each column. This structure indicates that the original independent components have been isolated, up to an arbitrary scalar multiplier and ordering.*

To quantify the divergence from this ideal structure, we utilize the Amari

Performance Index (Amari et al., 1996).

Definition 5.2 Amari Performance Index. *Let* $\mathbf{P} = \mathbf{B}_{est}\mathbf{A}$ *be the global transfer matrix. For a* $d \times d$ *matrix* $\mathbf{P}$ *with elements* p_{ij}*, the Amari error measures the structural divergence from a generalized permutation matrix:*

$$E(\mathbf{P}) = \frac{1}{2d}\sum_{i=1}^{d}\left(\sum_{j=1}^{d}\frac{|p_{ij}|}{\max_k |p_{ik}|} - 1\right) + \frac{1}{2d}\sum_{j=1}^{d}\left(\sum_{i=1}^{d}\frac{|p_{ij}|}{\max_k |p_{kj}|} - 1\right). \tag{5.1}$$

This formulation normalizes each row and column by their maximum absolute values. If $\mathbf{P}$ is a generalized permutation matrix, the index evaluates to zero. The experimental evaluations employ specific heuristic thresholds to interpret the Amari error as shown in Table 5.1.

Amari Error (E)	**Interpretation / Quality of Separation**
$E < 0.1$	Excellent to near-perfect source separation.
$0.1 \leq E < 0.3$	Good separation; the structural shape of the underlying sources is highly recoverable.
$0.3 \leq E < 0.5$	Moderate separation; some structural details may be lost.
$E \geq 0.5$	Degradation; multiple sources remain significantly mixed.
$E \geq 1.0$	The estimated matrix is an orthogonal projection with no meaningful relation to the true sources.

Table 5.1: *Heuristic thresholds for interpreting the Amari Performance Index.*

5.1.2 Computational Resource Regimes

Because OT-ICA relies on first-order gradient descent over the Orthogonal Group, its capacity to find the global optimum depends on computational resources, particularly random restarts. Conversely, FastICA utilizes a Newton-based fixed-point iteration. To ensure a strictly fair algorithmic comparison, FastICA is also provided with a substantial parameter of random initializations.

The high-dimensional tests in this chapter are evaluated under two standardized computational regimes as shown in Table 5.2.

Regime & Objective	OT-ICA Configuration	FastICA Configuration
Low Compute Baseline	**Restarts:** $\min(4 \times d, 150)$ **Deflation Phase:** 150 iterations **Symmetric Phase:** 200 iterations	**Restarts:** 50 **Max Iterations:** 1,000
High Compute Regime	**Restarts:** $\min(15 \times d, 600)$ **Deflation Phase:** 300 iterations **Symmetric Phase:** 500 iterations	**Restarts:** 50 **Max Iterations:** 5,000

Table 5.2: *Summary of the computational regimes. These iteration limits and restart thresholds separate standard optimization timeouts from mathematical convergence issues.*

5.2 OT-ICA Methodology Validation

We verify that OT-ICA recovers a mixture of non-Gaussian distributions in a low-dimensional environment. We generated $N = 10,000$ samples from four continuous independent sources: Laplace, Uniform, Student-t, and a Beta distribution parameterized at $(0.5, 0.5)$. These sources were linearly combined using a 4×4 mixing matrix $\mathbf{A}$.

The OT-ICA algorithm extracted the four structural signals from the mixture (Figure 5.1). Both global transfer matrices evaluate to generalized permutation matrices with Amari errors below the 0.1 threshold (Table 5.3), confirming OT-ICA is able to extract independent components as well as standard FastICA algorithm.

True Mixing Matrix (A)

$$\begin{bmatrix} 1.058 & 1.520 & -0.249 & 1.012 \\ 0.042 & -0.486 & 0.388 & -0.523 \\ -0.154 & -1.572 & -0.304 & -1.234 \\ -0.006 & 0.503 & 0.053 & 0.928 \end{bmatrix}$$

OT-ICA Global Transfer Matrix	**FastICA Global Transfer Matrix**
$\begin{bmatrix} -0.012 & -0.009 & -1.000 & -0.003 \\ 0.000 & 0.003 & -0.001 & 1.000 \\ -0.007 & 1.000 & 0.000 & 0.012 \\ -1.000 & -0.007 & 0.005 & -0.010 \end{bmatrix}$	$\begin{bmatrix} -1.000 & -0.012 & 0.003 & -0.008 \\ 0.012 & -1.000 & 0.001 & -0.004 \\ -0.002 & -0.010 & -0.001 & -1.000 \\ -0.010 & -0.010 & -1.000 & -0.001 \end{bmatrix}$

Amari Performance Index

OT-ICA Error: **0.0155** | FastICA Error: **0.0188**

Table 5.3: *The ground truth mixing matrix and the resulting global transfer matrices. Both algorithms yield matrices that closely approximate generalized permutation matrices.*

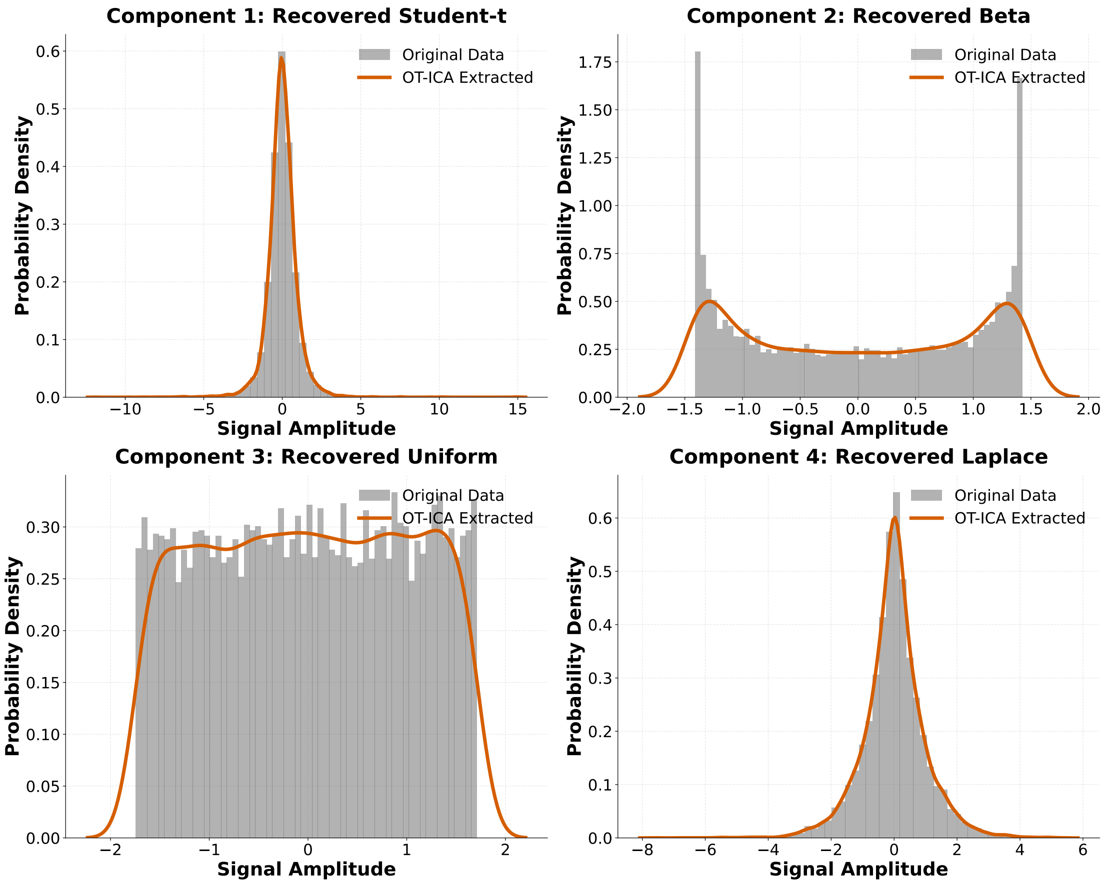

Figure 5.1: *Empirical probability distributions of the 4D baseline validation. Original sources (gray) are overlaid with the sources extracted by OT-ICA (orange).*

5.3 Heterogeneous Mixture Tests

To evaluate optimization convergence on heterogeneous signals, we tested generalized hybrid mixtures containing concurrent super-Gaussian, sub-Gaussian, continuous, and discrete properties.

We generated linear mixtures across dimensions with $N = 10,000$. One source was fixed as a standard Gaussian distribution. The remaining $(d-1)$ independent sources were randomly drawn from eight distributions: Laplace, Bernoulli, Uniform, Student-t, Poisson, Binomial, Chi-squared, and Exponential.

The empirical results (Figure 5.2) demonstrate that FastICA exhibits instability in three of the five hybrid environments: full hybrid, zero Gaussian, and discrete-only. In these cases, FastICA fails to converge ($E > 0.5$). Conversely, OT-ICA maintains separation ($E < 0.5$) across the dimensions for all but the strictly discrete environments.

To understand the divergence in performance observed in these hybrid environments, the subsequent sections isolate and evaluate the specific algorithmic shortcomings of both frameworks. Section 5.4 formally investigates the mathemat-

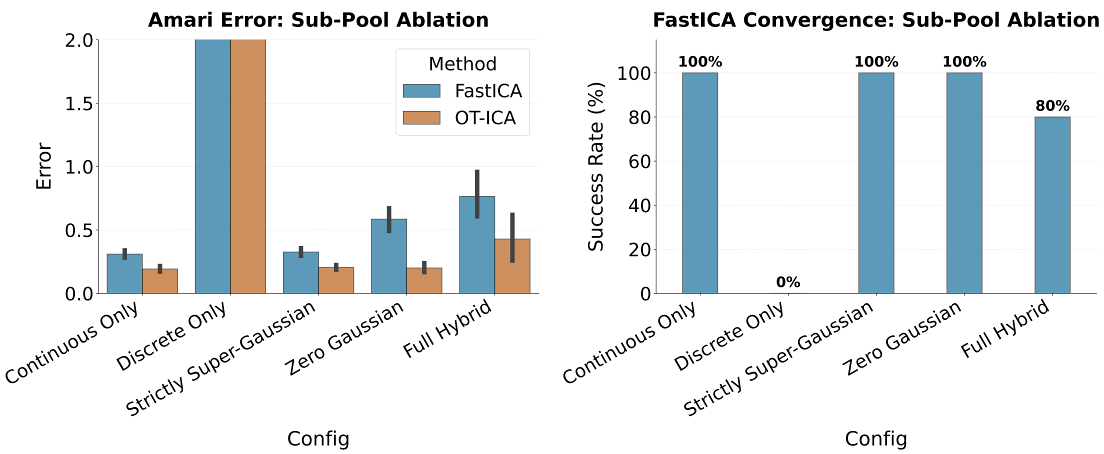

Figure 5.2: *Amari error comparison for the Generalized Hybrid Mixture. OT-ICA maintains stability across the tested dimensions.*

ical conditions under which FastICA's proxy contrast functions fail to converge. Following this, Section 5.5 isolates the discrete environments to analyze the contrast resolution and optimization bottlenecks for both FastICA and OT-ICA.

5.4 Algorithmic Limitations of FastICA

The instability of FastICA in hybrid mixtures suggests algorithmic limits to its objective function. We evaluate two specific limitations of FastICA's fixed-point iteration resulting from its reliance on proxy contrast functions.

5.4.1 The Zero Negentropy Condition

FastICA approximates negentropy using a non-quadratic contrast function $G(\cdot)$, such as the *logcosh* function. The approximation isolates non-Gaussianity via the difference of expectations:

$$\mathbb{E}[G(x)] - \mathbb{E}[G(\nu)] \tag{5.2}$$

where ν is a standard Gaussian random variable. If a latent independent source possesses a non-Gaussian distribution such that its expected value under the contrast function equals that of a Gaussian ($\mathbb{E}[G(x)] - \mathbb{E}[G(\nu)] = 0$), the approximated negentropy evaluates to zero. In this scenario, the objective function provides no gradient signal, and the algorithm fails to extract the source.

5.4.2 Empirical Results: Zero Negentropy

We constructed a symmetric Trimodal Gaussian Mixture parameterized by peak locations $\{-b, 0, b\}$ and variance σ^2. To engineer a zero-negentropy distribution,

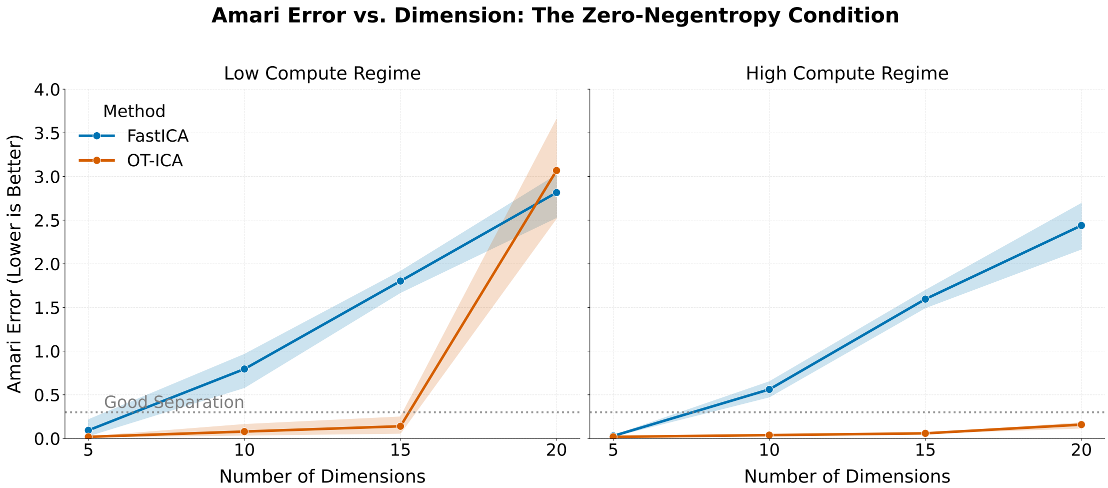

Figure 5.3: *Amari error comparison for the zero-negentropy distribution up to* $d = 20$.

we solved the following system of equations:

$$b^2 + \sigma^2 = 1 \quad \text{(Unit Variance Constraint)} \tag{5.3}$$

$$\int G(x)\text{PDF}(x)dx - \mathbb{E}[G(\nu)] = 0 \quad \text{(Zero Negentropy Constraint)} \tag{5.4}$$

This yields a distribution that is strictly non-Gaussian but evaluates to zero under the FastICA objective. We generated linear mixtures of this source across dimensions ($d \leq 20$) with $N = 10,000$ samples under both compute regimes.

The empirical results (Figure 5.3) detail the boundaries of this limitation. While FastICA, aided by random restarts, achieved separation ($E < 0.3$) at the lowest evaluated dimension ($d = 5$), it failed to converge for all dimensions $d > 5$, yielding high Amari errors ($E > 0.5$).

At $d = 10$, FastICA was allocated 50 parallel restarts compared to OT-ICA's 40 restarts in Low-Compute regime as defined in Table 5.2. But FastICA failed to converge, showing that the non-convergence is a limitation of the contrast function rather than a lack of optimization resources. In contrast, OT-ICA maintained separation ($E < 0.3$) up to $d = 15$ in the Low Compute regime and scaled successfully up to $d = 20$ in the High Compute regime.

5.4.3 The Vanishing Curvature Condition

FastICA utilizes a Newton (second-order) fixed-point iteration. The stability of this convergence relies on a condition governing the curvature of the optimization landscape (Hyvärinen et al., 2001, Chapter 8), (Appendix A.1).

Defining $g(x) = G'(x)$ and $g'(x) = G''(x)$, FastICA approximates the Hessian matrix $\mathbf{H}$ dynamically as a diagonal matrix. For a given component i, the diagonal entry scales according to the expectation:

$$\mathbf{H}_{ii} \approx \mathbb{E}\big[s_i g(s_i) - g'(s_i)\big] \tag{5.5}$$

The fixed-point Newton update requires applying the inverse of this Hessian ($\mathbf{H}^{-1}$). The inverse of the diagonal entry is formulated as:

$$\mathbf{H}_{ii}^{-1} \propto \frac{1}{\mathbb{E}\big[s_i g(s_i) - g'(s_i)\big]} \tag{5.6}$$

If a non-Gaussian source distribution causes the expectation in the denominator to evaluate to zero, the inverse becomes undefined. This analytical divide-by-zero error makes the update step intractable, preventing optimization convergence.

5.4.4 Empirical Results: Vanishing Curvature

We engineered a continuous Trimodal Gaussian Mixture source to trigger this condition. Setting $h(x) = x\tanh(x) - 1 + \tanh^2(x)$, we solved for parameters b and p satisfying the system:

$$\int x^2 \mathrm{PDF}(x)dx = 1 \tag{5.7}$$

$$\int h(x)\mathrm{PDF}(x)dx = 0 \tag{5.8}$$

This distribution is strictly non-Gaussian, but its curvature evaluates to zero (Figure 5.4).

We evaluated FastICA and OT-ICA across dimensions ($d \leq 20$) under both compute regimes.

The empirical results (Figure 5.5) detail the boundary of this limitation. FastICA successfully achieves source separation ($E < 0.3$) at the lowest evaluated dimension ($d = 5$) aided by its random restarts. However, it fails to converge for all dimensions $d \geq 10$, yielding high Amari errors ($E > 0.6$).

At $d = 10$, FastICA was allocated 50 parallel restarts compared to OT-ICA's 40 restarts. But FastICA failed to converge, demonstrating it's bad performance due to dependence on the inverse Hessian matrix in higher dimensions. OT-ICA, utilizing the L-BFGS Quasi-Newton solver, bypassed the Hessian requirement and successfully maintained separation ($E < 0.3$) through $d = 15$ and $d = 20$ in Low- and High-Compute regimes respectively.

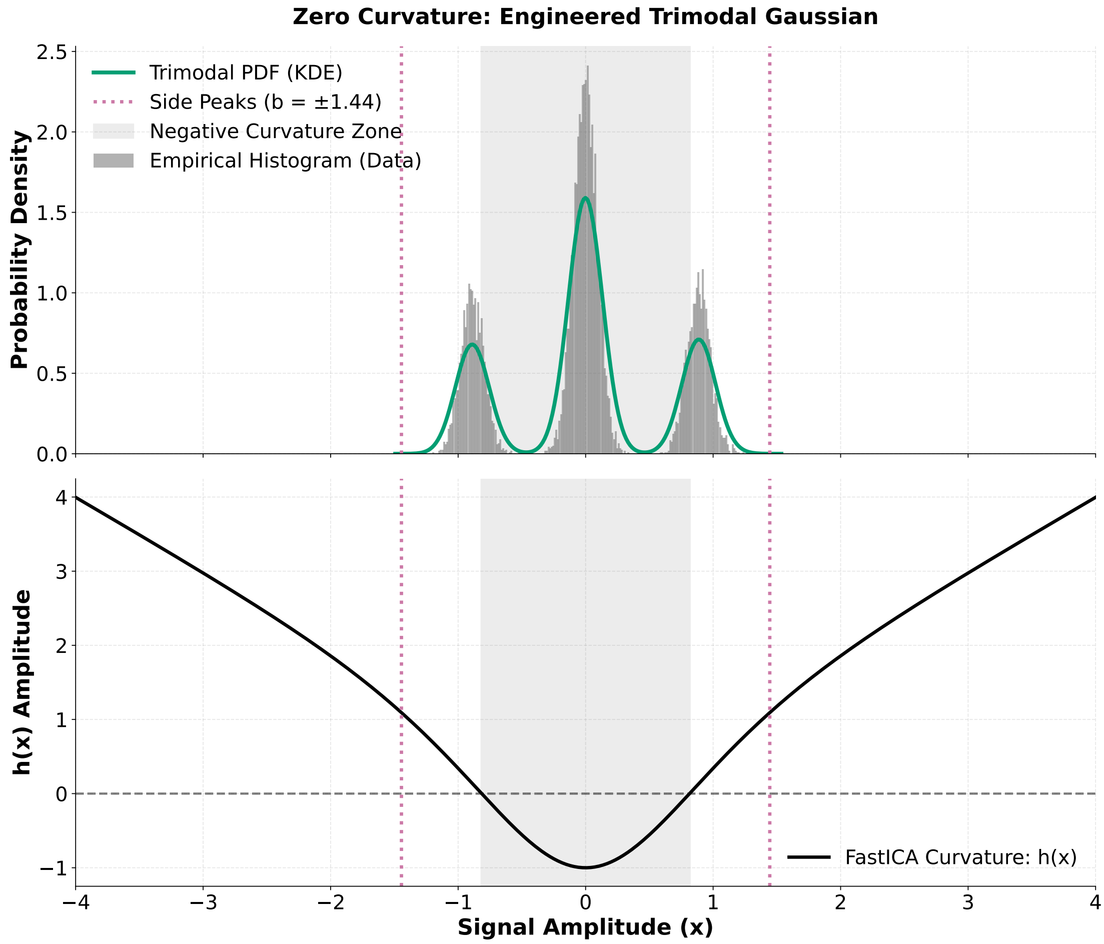

Figure 5.4: *The engineered Trimodal Gaussian Mixture. The central mass generates negative algorithmic curvature, while the side peaks generate positive algorithmic curvature. The parameters b and p are solved such that these regions cancel ($\int h(x)PDF(x)dx = 0$) while maintaining unit variance.*

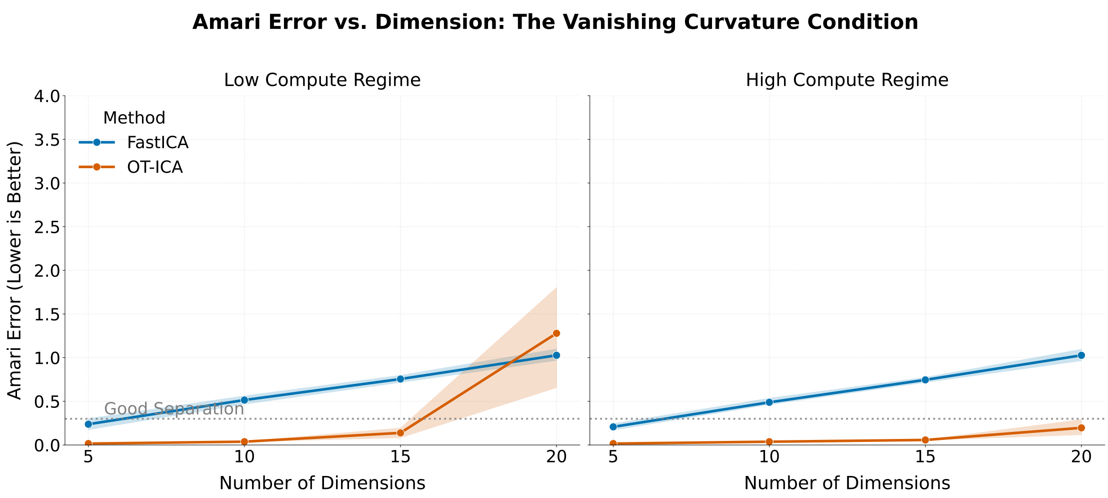

Figure 5.5: *Amari error comparison for the zero-curvature distribution up to $d = 20$.*

5.5 Optimization in Discrete Environments

While OT-ICA resolves continuous distributions efficiently, discrete data presents an optimization bottleneck, as observed in the generalized hybrid tests.

5.5.1 Empirical Results on Discrete Data

We isolated specific discrete distributions to observe their impact. We generated mixtures containing one Gaussian source and $(d-1)$ discrete sources.

Both algorithms report high Amari errors ($E > 1.0$) for multi-step count-based geometries such as standard Poisson ($\lambda = 3$) and Binomial ($n = 10$) distributions (Figure 5.6).

5.5.2 The Gradient Landscape of Discrete Distributions

We quantified the theoretical non-Gaussianity of these distributions to determine if the failure was due to the contrast metrics or to optimization complexity.

Distribution	W_2^2 **Distance**	**Logcosh Proxy Negentropy**
Laplace (Continuous Baseline)	0.0398	0.001214
Binomial (Standard, $n = 10$)	0.0344	0.000031
Poisson (Standard, $\lambda = 3.0$)	0.0513	0.000000
Binomial (Highly Non-Gaussian, $n = 2$)	0.2022	0.000267
Poisson (Highly Non-Gaussian, $\lambda = 0.5$)	0.3384	0.000085

Table 5.4: *Comparison of non-Gaussianity metrics across distributions.*

An evaluation of the metrics (Table 5.4) demonstrates that the squared Wasserstein distance provides orders of magnitude greater resolution for identifying

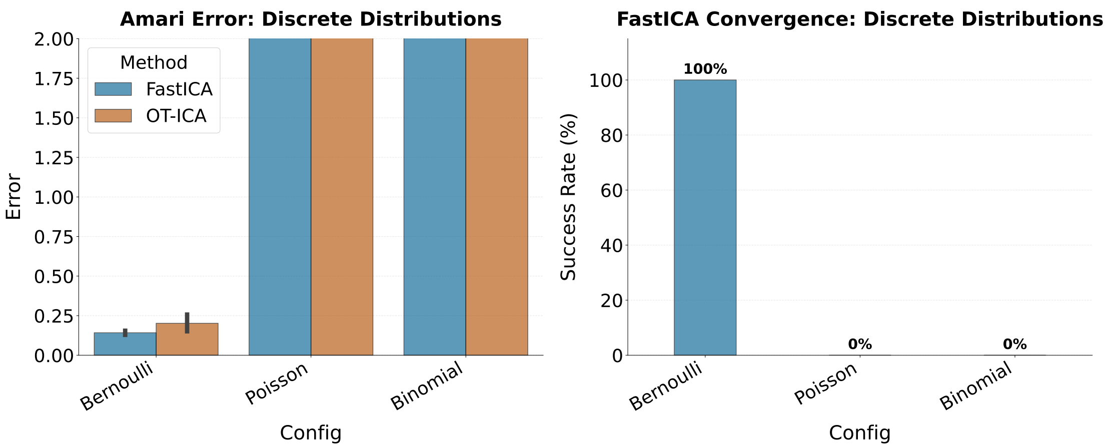

Figure 5.6: *Amari error for discrete mixtures. Both algorithms separate binary data (Bernoulli) but exhibit higher error on multi-step count data (Poisson, Binomial).*

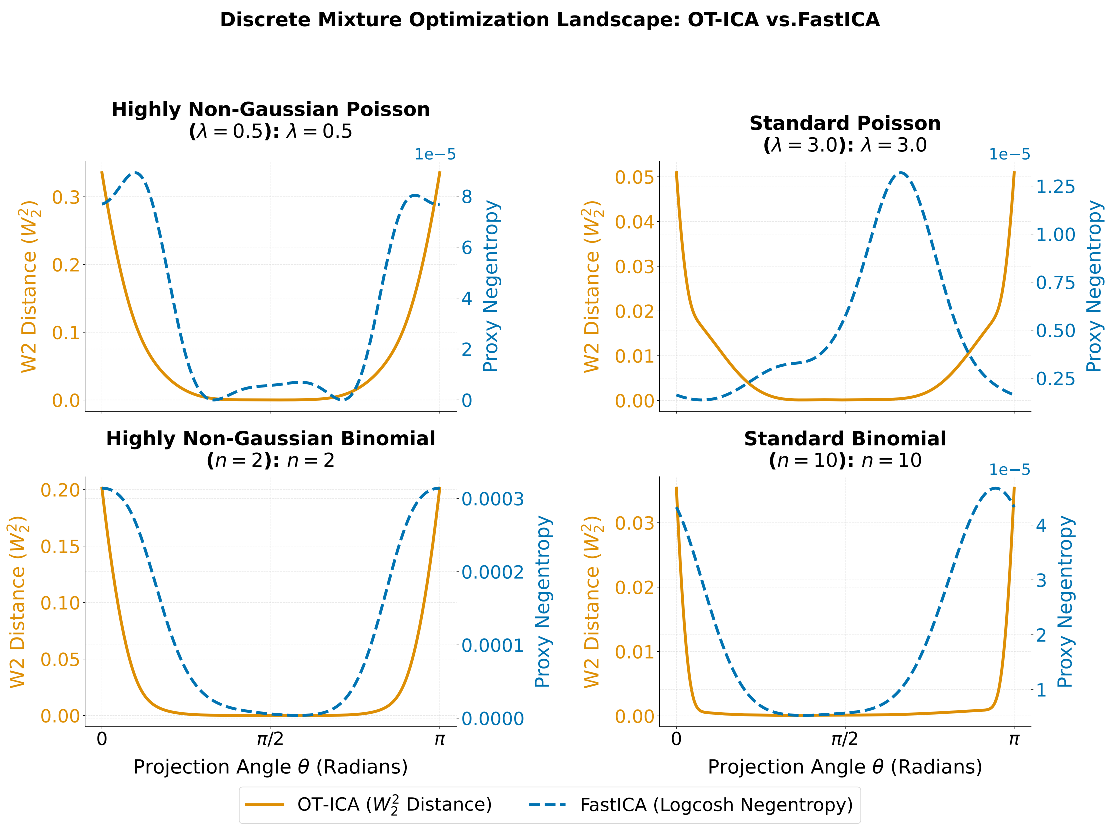

Figure 5.7: *The 1D optimization landscapes. The step-function geometry of discrete data creates flat plateaus where the OT-ICA gradient evaluates to zero.*

non-Gaussianity than FastICA's Logcosh approximation, which evaluates to near-zero scales for count-based data. The W_2^2 distance successfully measures higher levels of geometric divergence in highly non-Gaussian discrete parameterizations. This indicates that the failure to unmix discrete sources must not originate from a lack of contrast by the Wasserstein metric, rather the failure might originate from optimization landscape issues.

For OT-ICA, the geometric matching reflects the step-function nature of the discrete empirical CDF (Figure 5.7). This creates flat plateaus where the gradient is close to zero. This non-smooth landscape might stall the first-order L-BFGS solver, preventing optimization convergence despite the high-resolution contrast by the Wasserstein metric.

5.5.3 Empirical Results: Highly Non-Gaussian Parameterizations

We evaluated the algorithms on the skewed discrete parameterizations.

For the highly non-Gaussian Poisson mixture ($\lambda = 0.5$), FastICA did not converge (Figure 5.8). OT-ICA achieved separation ($E \approx 0.15$), as the skewed nature of the Poisson distribution creates a narrower plateau at the minimum of

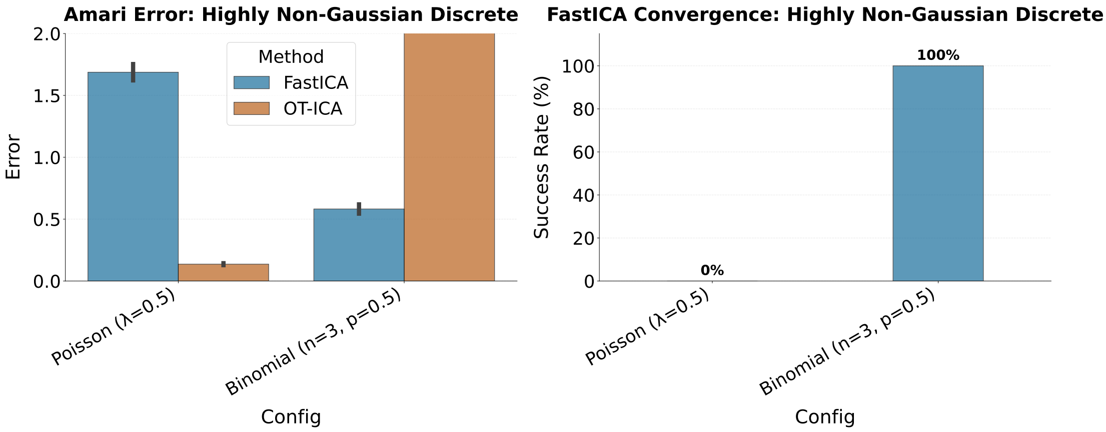

Figure 5.8: *Amari Error for skewed discrete parameterizations. OT-ICA achieves separation on the Poisson distribution but does not converge on the symmetric Binomial.*

the optimization surface. For the highly non-Gaussian Binomial mixture ($n = 2$), OT-ICA got stuck, resulting in an Amari error of $E > 2.0$, due to the wider plateau at the optimum.

5.6 Computational Scaling

To evaluate scaling efficiency, we benchmarked OT-ICA against FastICA using a linear mixture of continuous Laplace sources across increasing dimensions with a fixed sample size of $N = 10,000$.

While OT-ICA maintains error rates comparable to FastICA ($E < 0.3$) until $d = 40$ and $d = 55$ in the Low- and High-Compute regime respectively, both algorithms exhibit a loss in unmixing quality at $d = 40$ (Figure 5.9), where the Amari error exceeds the good separation ($E = 0.3$) threshold. This limit is a manifestation of the curse of dimensionality, where the finite sample size

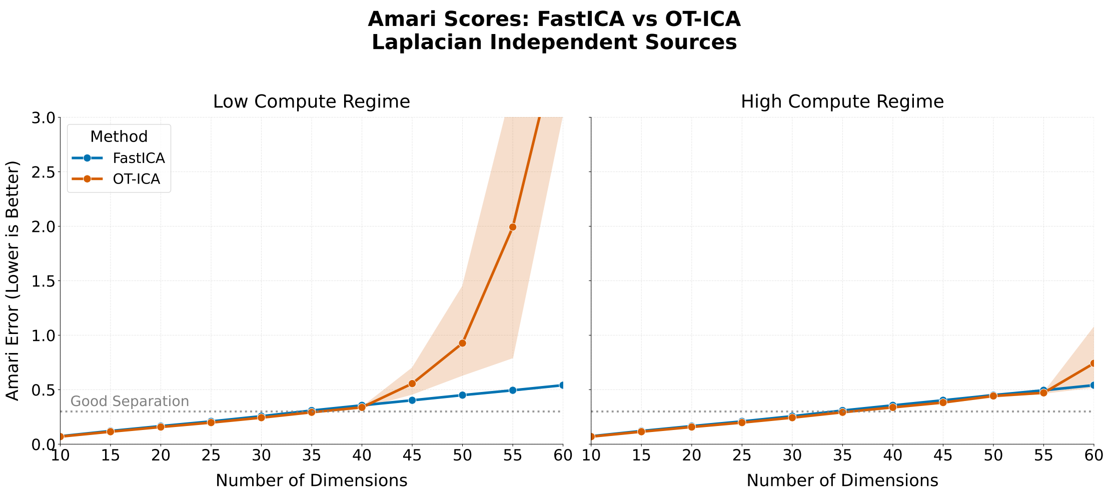

Figure 5.9: *Amari error comparison across dimensions for Laplacian sources.*

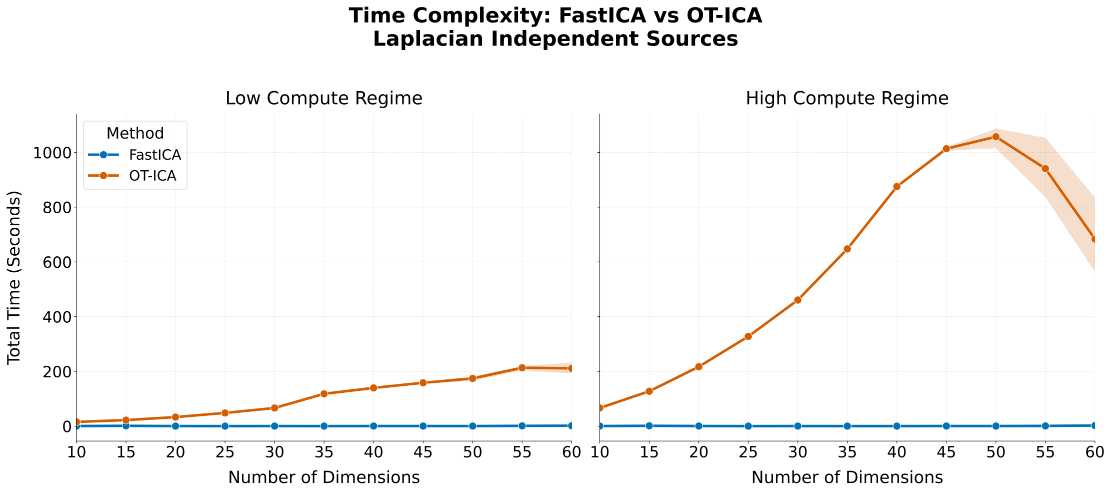

Figure 5.10: *Total execution time (in seconds) for FastICA and OT-ICA across dimensions.*

($N = 10,000$) fails to populate the state space densely enough to resolve rotations. A numerical example of this statistical noise ceiling is provided in Appendix A.3.

The execution time for OT-ICA scales more steeply than FastICA (Figure 5.10). The complexity of FastICA's Newton update evaluates to $\mathcal{O}(dN)$ per iteration. In contrast, as detailed in Section 4.3, OT-ICA incorporates the parallel sorting of projections across multiple restarts (K), resulting in a per-iteration complexity of $\mathcal{O}(K \cdot (dN \log N + d^3))$. Consequently, FastICA remains computationally better for exclusively homogeneous, high dimensional continuous mixtures. But OT-ICA provides reliable results for most mixture combinations other than some exclusively discrete mixtures.

5.7 The Case for OT-ICA

The evaluations clarify the trade-offs between optimization efficiency and result reliability. FastICA provides rapid $\mathcal{O}(dN)$ execution but relies on parametric proxy approximations that systematically fail on specific non-Gaussian structures, resulting in zero-negentropy and vanishing curvature limitations.

The empirical results demonstrate that the squared Wasserstein distance provides a high-resolution contrast for non-Gaussianity. Because OT-ICA maps the full empirical CDF geometry via this metric, it successfully isolates the independent sources.

While OT-ICA's first-order gradient evaluation over the Orthogonal Group incurs higher computational costs, and its reliance on empirical sorting creates optimization bottlenecks for strictly discrete distributions, these challenges originate from the current optimization pipeline rather than the theoretical validity of the metric itself. Potential algorithmic strategies to improve this optimization framework are discussed as future work in the concluding chapter. Despite these current computational trade-offs, OT-ICA provides a stable, mathematically sound framework for heterogeneous hybrid environments where FastICA fails.

5.8 Application: EEG Artifact Removal

In electroencephalography (EEG), the human skull acts as a linear volume conductor. Electrical signals generated by brain regions and ocular muscles (eye blinks) mix linearly before reaching the scalp electrodes. Because eye blinks are sparse and super-Gaussian, we apply the OT-ICA framework to isolate and remove them.

The raw frontal electrodes contain synchronized high-amplitude spikes (Figure 5.11, Plot A). The OT-ICA framework successfully isolated the ocular artifact into a single independent component (Comp 1), allowing for the analysis of the underlying brain waves.

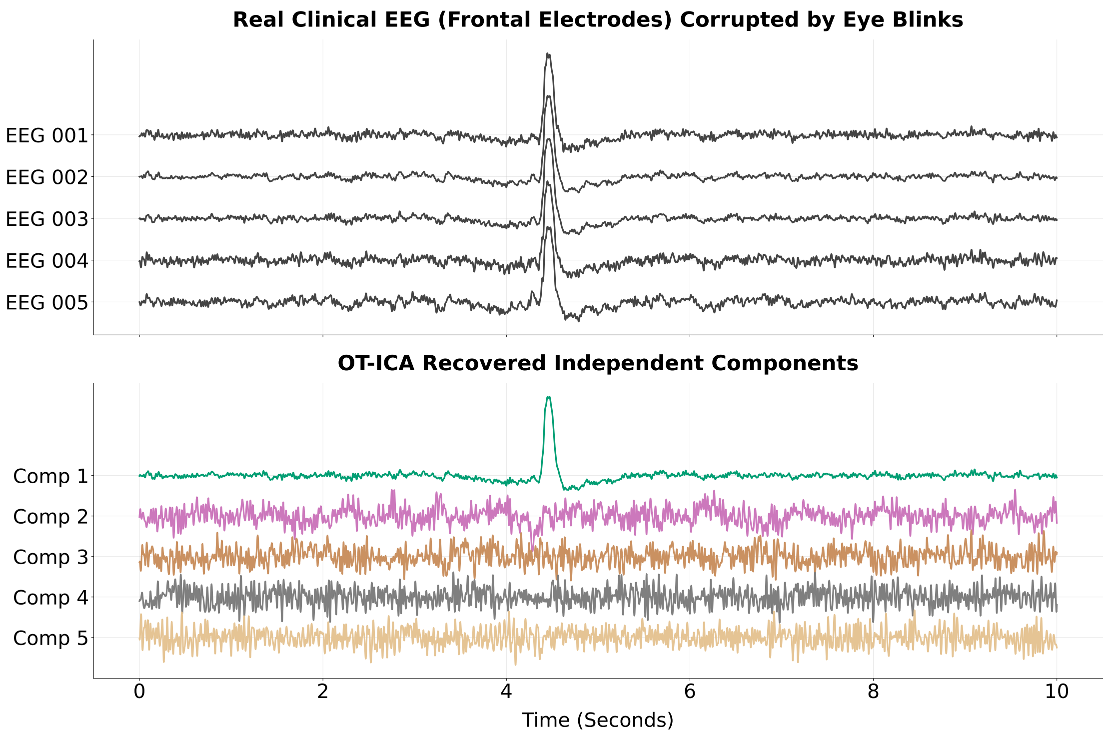

Figure 5.11: ***Top (Plot A):*** *Five frontal EEG channels containing eye-blink artifacts.* ***Bottom (Plot B):*** *The ocular artifact is isolated into a single component (Comp 1).*

5.9 Application: Price Discovery and Information Shares

In fragmented financial markets, an identical asset is often traded simultaneously across multiple venues. While short-term price changes across these markets may deviate due to microstructure noise, arbitrage ensures they are bound together by an unobserved "efficient price" random walk. A central problem in market microstructure is determining the Information Share (IS) (Hasbrouck, 1995; Baillie, Booth, Tse, & Zabotina, 2002), that is, quantifying the proportional contribution of each market to the variance of this fundamental price discovery process. Recently, Zema and Cordoni (Zema & Cordoni, 2025) demonstrated that this econometric identification problem can be uniquely resolved by leveraging the non-Gaussian properties of the underlying price innovations.

To model this cointegrated system, we adopt the standard Vector Error Correction Model (VECM) (Engle & Granger, 1987; Johansen, 1991). Let $P_t = [p_{1,t}, p_{2,t}, \ldots, p_{n,t}]^\top$ be the vector of logarithmic prices across n markets at time t. The price dynamics are formulated as:

$$\Delta P_t = \boldsymbol{\alpha}\boldsymbol{\beta}^\top P_{t-1} + \sum_{i=1}^{k} \boldsymbol{\Phi}_i \Delta P_{t-k} + u_t \tag{5.9}$$

where $\boldsymbol{\beta}$ contains the cointegrating vectors that enforce the long-term equilibrium between the markets, and $\boldsymbol{\alpha}$ represents the speed of adjustment to this equilibrium. The term u_t represents the reduced-form price innovations (residuals) with a contemporaneous covariance matrix $\boldsymbol{\Omega} = \mathbb{E}[u_t u_t^\top]$.

In order to measure the true Information Share of each venue, these reduced-form residuals u_t must be decomposed into mutually independent, structural market shocks ϵ_t (where $\mathbb{E}[\epsilon_t] = \mathbf{0}$ and $\mathrm{Cov}(\epsilon_t) = \mathbf{I}_n$). This decomposition requires identifying an instantaneous impact matrix $\mathbf{B}$ such that:

$$u_t = \mathbf{B}\epsilon_t \tag{5.10}$$

Expanded into full matrix notation, this relationship is expressed as:

$$\begin{bmatrix} u_{1,t} \\ u_{2,t} \\ \vdots \\ u_{n,t} \end{bmatrix} = \begin{bmatrix} b_{11} & b_{12} & \cdots & b_{1n} \\ b_{21} & b_{22} & \cdots & b_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ b_{n1} & b_{n2} & \cdots & b_{nn} \end{bmatrix} \begin{bmatrix} \epsilon_{1,t} \\ \epsilon_{2,t} \\ \vdots \\ \epsilon_{n,t} \end{bmatrix} \tag{5.11}$$

By substituting this relationship into the contemporaneous covariance matrix of the residuals, and utilizing the assumption that the structural shocks are mutually independent with unit variance ($\mathbb{E}[\epsilon_t \epsilon_t^\top] = \mathbf{I}_n$), the covariance constraint emerges in three steps:

$$\begin{aligned} \mathbf{\Omega} &= \mathbb{E}[u_t u_t^\top] \\ &= \mathbf{B}\, \mathbb{E}[\epsilon_t \epsilon_t^\top]\, \mathbf{B}^\top \\ &= \mathbf{B}\mathbf{I}_n\mathbf{B}^\top = \mathbf{B}\mathbf{B}^\top \end{aligned} \tag{5.12}$$

Because $\mathbf{\Omega}$ is symmetric, knowing the covariance matrix only partially identifies $\mathbf{B}$. Standard econometric approaches resolve this ambiguity using a Cholesky decomposition, forcing $\mathbf{B}$ to be a lower-triangular matrix. However, this imposes an arbitrary, recursive causal ordering on the markets, leading to different Information Share results depending on how the markets are ordered in the vector P_t.

To achieve a unique, order-invariant identification, any valid mixing matrix $\mathbf{B}$ can be factored into two components:

$$\mathbf{B} = \mathbf{S}\mathbf{C} \tag{5.13}$$

where $\mathbf{S}$ is an arbitrary whitening matrix satisfying $\mathbf{\Omega} = \mathbf{S}\mathbf{S}^\top$ (such as the Cholesky factor or an eigenvalue decomposition), and $\mathbf{C}$ is a strict orthogonal rotation matrix ($\mathbf{C}\mathbf{C}^\top = \mathbf{I}_n$).

By substituting this into our structural model, we get $u_t = \mathbf{S}\mathbf{C}\epsilon_t$, which can be rearranged as:

$$\mathbf{S}^{-1}u_t = \mathbf{C}\epsilon_t \tag{5.14}$$

The term $\mathbf{S}^{-1}u_t$ represents the whitened price innovations. Identifying the structural shocks ϵ_t now reduces to a search over the Orthogonal Group for the rotation matrix $\mathbf{C}$.

Classical models relying exclusively on Gaussian assumptions cannot identify this orthogonal rotation, as any orthogonal rotation of a spherically symmetric Gaussian vector yields another valid Gaussian vector. However, high-frequency financial market innovations exhibit highly non-Gaussian, heavy-tailed properties. By applying the OT-ICA framework to the whitened residuals, we leverage this non-Gaussianity to lock the orthogonal rotation $\mathbf{C}$, thereby recovering the true structural matrix $\mathbf{B}$ and enabling a unique, order-invariant calculation of the market Information Shares.

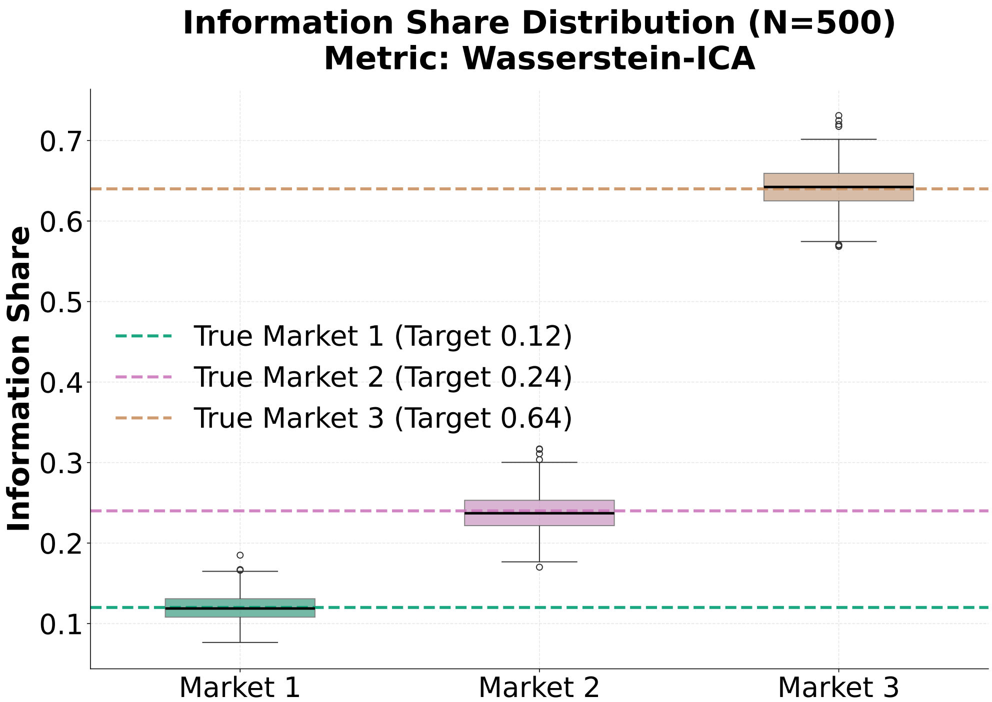

Figure 5.12: *Pairplot demonstrating the recovery of true structural shocks versus estimated shocks from OT-ICA for a single Monte Carlo run. The linear alignment indicates the successful unmixing of the latent market innovations.*

5.9.1 Empirical Results on Simulated Market Data

We simulated a fragmented scenario targeting three markets with pre-defined Information Shares of 12%, 24%, and 64%. Market innovations were generated to satisfy the non-Gaussianity requirements, and we executed 500 Monte Carlo simulations comprising 5,000 observations each.

A representative run demonstrates the correlation between the true generative shocks and the empirical shocks estimated by OT-ICA (Figure 5.12).

By mapping the statistical components to their corresponding markets, we calculated the empirical information shares. The estimated information shares converge closely to their theoretical targets across all 500 Monte Carlo simulations (Table 5.5).

Market / Source	True IS	Estimated IS (Mean)	Standard Deviation
Market 1	0.1200	0.1212	0.0161
Market 2	0.2400	0.2399	0.0245
Market 3	0.6400	0.6389	0.0260

Table 5.5: *Results comparing true Information Shares (IS) against the estimated IS recovered via OT-ICA over 500 runs.*

The empirical density of the estimated information shares confirms that the

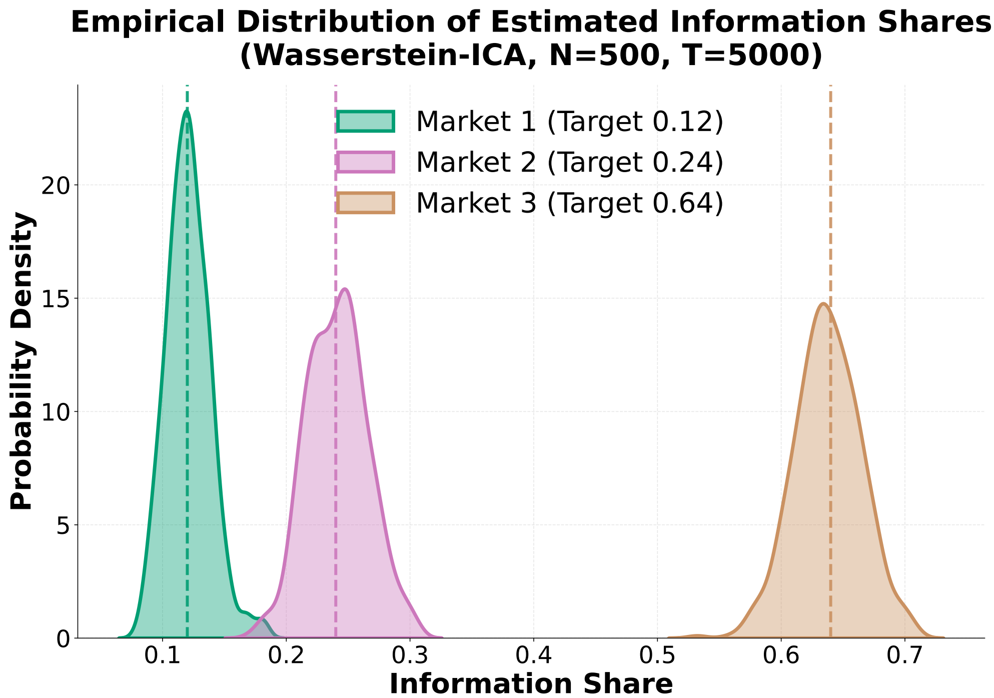

Figure 5.13: *Empirical density distributions of the estimated Information Shares across 500 Monte Carlo simulations. The vertical dashed lines represent the true theoretical values.*

estimations are tightly centered around the true parameter values across the simulation runs (Figure 5.13).

Thus, we conclude that by exploiting non-Gaussianity to resolve the orthogonal ambiguities inherent in standard VECMs, OT-ICA provides a reliable framework for measuring price discovery across markets.

Chapter 6

Conclusion

6.1 Summary of Contributions

Independent Component Analysis (ICA) remains a fundamental problem in unsupervised learning and signal processing. From an information geometry perspective, traditional approaches utilize mathematically sound information-theoretic measures, such as Kullback-Leibler divergence or negentropy, to achieve source separation. However, in algorithmic practice, solvers evaluate these measures using local parametric approximations and proxy contrast functions. This thesis was motivated by the application of the squared Wasserstein distance (W_2^2) as an actual distance metric. By directly minimizing the optimal transport cost to a standard normal distribution over the Orthogonal Group, the OT-ICA framework evaluates non-Gaussianity strictly through it's distance metric, bypassing the contrast and local density approximations.

The empirical evaluations quantified the advantages of this direct metric formulation. Newton-based proxy algorithms, such as FastICA, exhibited optimization non-convergence when presented with specific non-Gaussian topologies, including zero-negentropy distributions and optimization landscapes with vanishing curvature. By mapping the full empirical cumulative distribution function, OT-ICA resolved these constrained sources and maintained stable separation across generalized hybrid mixtures. To ensure computational viability, the framework integrated analytical Gaussian targets, batched vectorization, and Gaussian dithering techniques. The algorithm was further applied to real-world data, successfully isolating ocular artifacts in clinical EEG recordings and identifying orthogonal shocks for econometric price discovery.

Despite these advantages, the methodology is subject to specific computational and statistical limitations. Because computing the Hessian of the Wasserstein metric requires continuous density estimation, OT-ICA utilizes first-order Quasi-

Newton optimization with parallel random restarts, resulting in execution times that scale steeply compared to fixed-point solvers. The evaluated ICA frameworks are also bounded by the curse of dimensionality. A fixed finite sample size fails to densely populate high-dimensional spaces, establishing a statistical noise ceiling that degrades unmixing performance at higher dimensions. Finally, while Gaussian dithering smooths discrete inputs, highly symmetric and sparse discrete distributions generate zero-gradient plateaus that stall first-order solvers.

6.2 Future Work

Future research directions include the integration of entropic regularization for Sinkhorn distances (Cuturi, 2013; Peyré & Cuturi, 2019) to replace the Gaussian dithering step. Sinkhorn divergence provides a naturally differentiable approximation of the optimal transport metric that processes discrete distributions without geometrically altering the underlying data.

Improving computational scaling represents a primary objective. Integrating density approximation techniques, such as Kernel Density Estimation (KDE), could enable the continuous calculation of analytical Hessians, permitting rapid second-order optimization over the Orthogonal Group. Furthermore, leveraging Amortized Optimal Transport to learn the transport map offline would bypass the central iterative bottlenecks of the forward pass, specifically the $\mathcal{O}(d^2N)$ matrix projection operations and the $\mathcal{O}(dN \log N)$ empirical sorting requirements during runtime.

Integrating the OT-ICA framework into causal discovery offers a promising avenue. Causal structures are frequently modeled using Linear Non-Gaussian Acyclic Models (LiNGAM) (Shimizu, Hoyer, Hyvärinen, & Kerminen, 2006), which rely directly on ICA to identify the independent error terms that dictate causal directionality. Applying an actual metric, the Wasserstein distance, like OT-ICA could improve causal estimation in heterogeneous datasets where proxy solvers fail. The emerging field of Causal Representation Learning also seeks to infer causally related latent variables. The recently proposed Causal Component Analysis (CauCA) (Liang et al., 2023) bridges ICA and causal learning by modeling causal dependence among latent components. Integrating the robust Wasserstein metric into the CauCA framework could enhance the estimation of unmixing functions and causal mechanisms.

Finally, extending the geometric properties of optimal transport to nonlinear Independent Component Analysis represents a compelling frontier. Unsupervised learning of latent variable models via nonlinear ICA requires strict constraints on the function class to ensure identifiability (Buchholz, Besserve, & Schölkopf, 2022).

The optimal transport metric, which evaluates global structural deformations, may serve to effectively regularize these constrained function spaces, enabling reliable unmixing of nonlinear representations.

Appendix A

Mathematical Proofs and Derivations

A.1 Derivation of the FastICA Vanishing Curvature Condition

Proof of Theorem 8.1 (Hyvärinen et al., 2001, Chapter 8) Denote by $H(\mathbf{b})$ the function to be minimized/maximized, $E\{G(\mathbf{b}^T\mathbf{z})\}$. Make the orthogonal change of coordinates $\mathbf{r} = \mathbf{A}^T\mathbf{Q}^T\mathbf{b}$. Then we can calculate the gradient as $\frac{\partial H(\mathbf{r})}{\partial \mathbf{r}} = E\{\mathbf{s}g(\mathbf{r}^T\mathbf{s})\}$ and the Hessian as $\frac{\partial^2 H(\mathbf{r})}{\partial \mathbf{r}^2} = E\{\mathbf{s}\mathbf{s}^T g'(\mathbf{r}^T\mathbf{s})\}$. Without loss of generality, it is enough to analyze the stability of the point $\mathbf{r} = \mathbf{e}_1$, where $\mathbf{e}_1 = (1, 0, 0, 0, \dots)$. Evaluating the gradient and the Hessian at point $\mathbf{r} = \mathbf{e}_1$, we get using the independence of the s_i,

$$\frac{\partial H(\mathbf{e}_1)}{\partial \mathbf{r}} = \mathbf{e}_1 E\{s_1 g(s_1)\} \tag{A.1}$$

and

$$\frac{\partial^2 H(\mathbf{e}_1)}{\partial \mathbf{r}^2} = \mathrm{diag}(E\{s_1^2 g'(s_1)\}, E\{g'(s_1)\}, E\{g'(s_1)\}, \dots). \tag{A.2}$$

Making a small perturbation $\epsilon = (\epsilon_1, \epsilon_2, \dots)$, we obtain

$$\begin{aligned} H(\mathbf{e}_1 + \epsilon) &= H(\mathbf{e}_1) + \epsilon^T \frac{\partial H(\mathbf{e}_1)}{\partial \mathbf{r}} + \frac{1}{2}\epsilon^T \frac{\partial^2 H(\mathbf{e}_1)}{\partial \mathbf{r}^2}\epsilon + o(\|\epsilon\|^2) \\ &= H(\mathbf{e}_1) + E\{s_1 g(s_1)\}\epsilon_1 + \frac{1}{2}\Big[E\{s_1^2 g'(s_1)\}\epsilon_1^2 + E\{g'(s_1)\}\sum_{i>1}\epsilon_i^2\Big] + o(\|\epsilon\|^2) \end{aligned} \tag{A.3}$$

Due to the constraint $\|\mathbf{b}\| = 1$ we get $\epsilon_1 = \sqrt{1 - \epsilon_2^2 - \epsilon_3^2 - \dots} - 1$. Due to the fact that $\sqrt{1-\gamma} = 1 - \gamma/2 + o(\gamma)$, the term of order ϵ_1^2 is $o(\|\epsilon\|^2)$, i.e., of higher order, and can be neglected. Using the aforementioned first-order approximation for ϵ_1

we obtain $\epsilon_1 = -\sum_{i>1} \epsilon_i^2/2 + o(\|\epsilon\|^2)$, which finally gives

$$H(\mathbf{e}_1 + \epsilon) = H(\mathbf{e}_1) + \frac{1}{2}\Big[E\{g'(s_1) - s_1 g(s_1)\}\Big] \sum_{i>1} \epsilon_i^2 + o(\|\epsilon\|^2) \qquad \text{(A.4)}$$

which clearly proves $\mathbf{r} = \mathbf{e}_1$ is an extremum, and of the type implied by the condition of the theorem.

Proof of convergence of FastICA (Hyvärinen et al., 2001, Chapter 8) The convergence is proven under the assumptions that first, the data follows the ICA data model and second, that the expectations are evaluated exactly.

Let g be the nonlinearity used in the algorithm. In the case of the kurtosis-based algorithm, this is the cubic function, so we obtain that algorithm as a special case of the following proof for a general g. We must also make the following technical assumption:

$$E\{s_i g(s_i) - g'(s_i)\} \neq 0, \quad \text{for any } i \qquad \text{(A.5)}$$

which can be considered a generalization of the condition valid when we use kurtosis, that the kurtosis of the independent components must be nonzero. If this is true for a subset of independent components, we can estimate just those independent components.

To begin with, make the change of variable $\mathbf{r} = \mathbf{A}^T\mathbf{Q}^T\mathbf{b}$, as earlier, and assume that $\mathbf{r}$ is in the neighborhood of a solution (say, $r_1 \approx 1$ as before). As shown in the proof of Theorem 8.1, the change in r_1 is then of a lower order than the change in the other coordinates, due to the constraint $\|\mathbf{r}\| = 1$. Then we can expand the terms using a Taylor approximation for g and g', first obtaining

$$\begin{aligned} g(\mathbf{r}^T\mathbf{s}) = g(r_1 s_1) &+ g'(r_1 s_1)\mathbf{r}_{-1}^T\mathbf{s}_{-1} + \frac{1}{2}g''(r_1 s_1)(\mathbf{r}_{-1}^T\mathbf{s}_{-1})^2 \\ &+ \frac{1}{6}g'''(r_1 s_1)(\mathbf{r}_{-1}^T\mathbf{s}_{-1})^3 + O(\|\mathbf{r}_{-1}\|^4) \end{aligned} \qquad \text{(A.6)}$$

and then

$$\begin{aligned} g'(\mathbf{r}^T\mathbf{s}) = g'(r_1 s_1) &+ g''(r_1 s_1)\mathbf{r}_{-1}^T\mathbf{s}_{-1} \\ &+ \frac{1}{2}g'''(r_1 s_1)(\mathbf{r}_{-1}^T\mathbf{s}_{-1})^2 + O(\|\mathbf{r}_{-1}\|^3) \end{aligned} \qquad \text{(A.7)}$$

where $\mathbf{r}_{-1}$ and $\mathbf{s}_{-1}$ are the vectors $\mathbf{r}$ and $\mathbf{s}$ without their first components. Denote by $\mathbf{r}^+$ the new value of $\mathbf{r}$ (after one iteration). Thus we obtain, using the independence of the s_i and doing some tedious but straightforward algebraic manipulations,

$$r_1^+ = E\{s_1 g(r_1 s_1) - g'(r_1 s_1)\} + O(\|\mathbf{r}_{-1}\|^2) \qquad \text{(A.8)}$$

$$r_i^+ = \frac{1}{2}E\{s_i^3\}E\{g''(s_1)\}r_i^2 + \frac{1}{6}\text{kurt}(s_i)E\{g'''(s_1)\}r_i^3 + O(\|\mathbf{r}_{-1}\|^4), \quad \text{for } i > 1 \tag{A.9}$$

We obtain also

$$\mathbf{r}^* = \mathbf{r}^+/\|\mathbf{r}^+\| \tag{A.10}$$

This shows clearly that under assumption (A.5), the algorithm converges (locally) to such a vector $\mathbf{r}$ that $r_1 = \pm 1$ and $r_i = 0$ for $i > 1$. This means that $\mathbf{b} = ((\mathbf{QA})^T)^{-1}\mathbf{r}$ converges, up to the sign, to one of the rows of the inverse of the whitened mixing matrix $\mathbf{QA}$, which implies that $\mathbf{b}^T\mathbf{z}$ converges to one of the s_i. Moreover, if $E\{g''(s_1)\} = 0$, i.e., if the s_i has a symmetric distribution, as is usually the case, the convergence is cubic. In other cases, the convergence is quadratic. If kurtosis is used, however, we always have $E\{g''(s_1)\} = 0$ and thus cubic convergence. In addition, if $G(y) = y^4$, the local approximations are accurate, and the convergence is global.

A.2 Intractability of Newton Updates for Empirical OT-ICA

Theorem A.1 Intractability of Newton Updates. *For an empirical distribution mapped to a continuous target via sorting, computing a closed-form Hessian $\mathbf{H} = \nabla^2_{\mathbf{b}} W_2^2$ is analytically intractable without introducing external density estimation mechanisms. The Map Derivative $\nabla_{\mathbf{b}} T_{\mathbf{b}}$ relies upon the rate of change of cumulative probability mass, which strictly requires the continuous Probability Density Function (PDF) evaluated at the quantile boundaries.*

Proof. To implement a Newton step, the Hessian matrix of the squared Wasserstein distance, $\mathbf{H} = \nabla^2_{\mathbf{b}} W_2^2$, is required. We first define the objective function. Let the 1D projection of the data be $y = \mathbf{b}^\top \mathbf{X}$, and let $T_{\mathbf{b}}$ be the optimal transport map pushing this empirical projection to the target Gaussian distribution. The squared Wasserstein distance is defined as:

$$W_2^2(\mathbf{b}) = \mathbb{E}\Big[\big(\mathbf{b}^\top \mathbf{X} - T_{\mathbf{b}}(\mathbf{b}^\top \mathbf{X})\big)^2\Big]. \tag{A.11}$$

We compute the first differential (the gradient) of the objective with respect to the weight vector $\mathbf{b}$. Utilizing the envelope theorem for optimal transport (Santambrogio, 2015, Section 7.2), the variation of the optimal map evaluates to zero within the cost function, allowing us to differentiate the squared cost directly without differentiating the map itself at this first stage:

$$\mathbf{g} = \nabla_{\mathbf{b}} \left(\frac{1}{2} W_2^2 \right) = \frac{1}{2} \mathbb{E}\Big[2\big(\mathbf{b}^\top \mathbf{X} - T_{\mathbf{b}}(\mathbf{b}^\top \mathbf{X})\big) \nabla_{\mathbf{b}} (\mathbf{b}^\top \mathbf{X}) \Big]$$
$$= \mathbb{E}\Big[\mathbf{X}\big(\mathbf{b}^\top \mathbf{X} - T_{\mathbf{b}}(\mathbf{b}^\top \mathbf{X})\big) \Big]. \tag{A.12}$$

To compute the second differential (the Hessian $\mathbf{H}$), we must now differentiate the gradient vector $\mathbf{g}$ with respect to $\mathbf{b}$. Applying the derivative yields:

$$\mathbf{H} = \nabla_{\mathbf{b}} \mathbf{g} = \mathbb{E}\Big[\mathbf{X} \nabla_{\mathbf{b}} \big(\mathbf{b}^\top \mathbf{X} - T_{\mathbf{b}}(\mathbf{b}^\top \mathbf{X})\big)^\top \Big]$$
$$= \mathbb{E}\Big[\mathbf{X}\mathbf{X}^\top - \mathbf{X} \nabla_{\mathbf{b}} [T_{\mathbf{b}}(\mathbf{b}^\top \mathbf{X})]^\top \Big]. \tag{A.13}$$

The mathematical intractability lies entirely within the term $\nabla_{\mathbf{b}}[T_{\mathbf{b}}(\mathbf{b}^\top \mathbf{X})]$. Applying the total derivative using the chain rule yields two distinct components:

$$\nabla_{\mathbf{b}}[T_{\mathbf{b}}(\mathbf{b}^\top \mathbf{X})] = \underbrace{T'_{\mathbf{b}}(\mathbf{b}^\top \mathbf{X})\mathbf{X}}_{\text{Argument Derivative}} + \underbrace{(\nabla_{\mathbf{b}} T_{\mathbf{b}})(\mathbf{b}^\top \mathbf{X})}_{\text{Map Derivative}}. \tag{A.14}$$

By Caffarelli's regularity theory (Caffarelli, 1992), the transport map $T_{\mathbf{b}}$ is continuously differentiable, guaranteeing the existence of the Argument Derivative. The fundamental limitation lies within the Map Derivative.

In the 1D optimal transport setting, the transport map to a standard Gaussian relies on the empirical cumulative distribution function, $F_{\mathbf{b}}$, of the projected data $y = \mathbf{b}^\top \mathbf{X}$. Expanding the Map Derivative yields:

$$(\nabla_{\mathbf{b}} T_{\mathbf{b}})(y) = \nabla_{\mathbf{b}} \big[\Phi^{-1}(F_{\mathbf{b}}(y)) \big] = \frac{1}{\phi(\Phi^{-1}(F_{\mathbf{b}}(y)))} \nabla_{\mathbf{b}} F_{\mathbf{b}}(y). \tag{A.15}$$

The term $\nabla_{\mathbf{b}} F_{\mathbf{b}}(y)$ represents the rate of change of the cumulative probability mass as the projection plane $\mathbf{b}$ is rotated. The derivative of a cumulative distribution function's boundary strictly requires the underlying continuous probability density function (PDF) evaluated at that specific boundary. For empirical distributions consisting of discrete sampled points, this density does not natively exist, rendering the closed-form calculation of the Hessian analytically intractable without explicit external density estimation. ■

A.3 Statistical Limits of High-Dimensional Source Separation

In Section 5.6, the empirical results demonstrate that at $d = 40$ with a sample size of $N = 10,000$, both OT-ICA and FastICA experience a loss in unmixing quality, failing to reach an Amari error below 0.3. This structural plateau is a manifestation of the curse of dimensionality.

According to Fournier and Guillin (2015), the expected error of the empirical Wasserstein distance scales as a function of sample size and dimension:

$$\mathbb{E}[W_2(P_N, P)] \sim \mathcal{O}(N^{-1/d}) \tag{A.16}$$

This formulation provides a measure of the sparsity of points within a high-dimensional space, effectively estimating the proportional distance between nearest neighbor data points. Applying the fixed sample size of $N = 10,000$ to the evaluated dimensions yields the proportional noise floor (Table A.1).

Dimension (d)	**Proportional Distance Floor ($\sim (10^{-4})^{1/d}$)**
$d = 5$	0.158
$d = 10$	0.398
$d = 40$	0.794
$d = 50$	0.832

Table A.1: *Theoretical scaling of the nearest neighbor distance as dimensionality increases for a fixed sample size of $N = 10,000$.*

While this scaling law is not an absolute mathematical proof of algorithmic failure, it serves as a strong numerical indication of the statistical loss of resolution in the joint distribution. Because the input data is whitened (enforcing unit variance), the probability mass is generally centered and bounded. As dimensionality approaches $d = 40$, the expected distance between any two nearest neighbor data points scales up to ≈ 0.8.

In a standardized space, data points separated by a distance of 0.8 are severely isolated. The empirical density degenerates into massive, empty regions. Consequently, the optimization gradient generated by matching the target distributions is swallowed by this sparsity. It becomes unreasonable for any continuous gradient or fixed-point solver to mathematically differentiate between subtle rotational improvements on the manifold, resulting in the unmixing matrix stalling at a structural approximation bound, manifesting empirically as the 0.3 Amari error plateau.

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